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**(U) Remote Chemical Sensor
Development – CB10674: Concepts and
Technology Assessment for Leave-in-Place
Chemical Sensors**

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RESEARCH AND TECHNOLOGY DIRECTORATE

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(U) EXECUTIVE SUMMARY

(U) This report provides performance characteristics and user feedback on conceptual leave-in-place chemical sensors (LIPCS), identifies commercial off-the-shelf technologies that are sufficiently mature to address the envisioned operational applications, and highlights relevant emerging science and technology. As designed, an LIPCS is a small, inexpensive, “disposable” sensor that is deployable by various means and in different situations, to indicate the presence of chemical vapor hazards without increasing a Warfighter’s physical and cognitive burden. Conceptual development of deployment was investigated through interviews with operational users from the U.S. Army, U.S. Marine Corps, U.S. Navy, U.S. Air Force, and other interested stakeholders, over a six-month period. The authors of this report also attended the 2019 Chemical Biological Operational Analysis event (held at Camp Dawson, WV) and presented in the Concept Feedback Tent. Over a three-day period, more than 70 users, operators, and attendees were interviewed to acquire real-time feedback from daily field exercises. The principal question was how an LIPCS-type device could have been employed to mitigate chemical hazards during the exercise.

(U) Conceptual LIPCS devices were described to the users as being intended for chemical hazard monitoring or reconnaissance, disposable (no larger than a single round of ammunition, up to a cartridge size), and deployable during missions by way of carrying, tossing, depositing, or launching, such as along routes of advance or egress. The sensors would be small enough to allow for multiple LIPCSs to be carried by a single individual, and they would be monitored remotely, so that a user would not have to physically interact with the LIPCS device.

(U) Combining this LIPCS description with input from the various user discussions, five main operational scenarios and applications of interest were determined in which LIPCSs would enhance situational awareness of the threat environment:

- (U//FOUO) unattended perimeter-monitor chemical sensor deployment in choke points and for route surveillance, as well as clearance, reconnaissance, egress, and ingress points;
- (U//FOUO) subterranean, room, and hallway reconnaissance to provide the ability to deploy the sensor without physically breaching the area of interest;
- (U//FOUO) deployment of LIPCSs around named areas of interest and target areas of interest for surveillance and reconnaissance (“outside the wire” placement), to include deployment by air-dropping the devices using unmanned aerial systems;
- (U//FOUO) deployment of sensors around airfields, logistics support areas, and ports to continuously monitor for vapor hazards and surface contamination; and
- (U//FOUO) deployment to establish safe areas around structurally unstable or collapsed infrastructure.

(U//FOUO) These five scenarios identify a desire for small-size, low-weight, and low-power sensors that can be deployed in areas of interest to remotely monitor chemical vapor threats,

(U//FOUO)

which is a technology that does not exist. A list of LIPCS performance metrics were identified that would be beneficial to fulfill these operational scenarios:

- (U//FOUO) LIPCS devices should be ~1 lb or less (optimally, less than 1/3 lb).
- (U//FOUO) Sensor distance to operator should be sufficient to enable detect to warn (masking up).
- (U//FOUO) Different modes of alarm operation should be used, such as silent (transmits to listening or observational post), audible, or visible (to be read by an external device).
- (U//FOUO) LIPCSs should be easily deployed and operated by either chemical, biological, radiological, and nuclear (CBRN) specialists or general-purpose forces.
- (U//FOUO) Battery life should be 8–12 h with a goal of 24 h or greater.
- (U//FOUO) Anomaly hazard detection should be a threshold requirement; threat classification should be an objective.
- (U//FOUO) LIPCSs should be able to connect to existing or developing tactical information systems.

(U) During discussions about the potential desired functionalities of LIPCS, users stated, “It is less important to identify the threat than to classify the presence of a threat.” In general, the LIPCS is expected to be a less specific, potentially less sensitive sensing and alerting capability than the standard Joint Chemical Agent Detector. When deployed as a detect-to-warn based functionality, the LIPCS should indicate anomalous circumstances that are consistent with a threat scenario and perhaps broadly classify the situation as a possible chemical attack.

(U) Following concept development of the use cases for LIPCSs, a survey of current technologies was performed to identify developing and advanced sensor solutions that could be assessed or matured through science and technology strategies. Four technological areas were identified and are discussed in greater detail in Section 4:

- (U//FOUO) impedance techniques (graphene, metal–organic frameworks, molecular imprinted polymers, and functionalized carbon nanotubes);
- (U//FOUO) colorimetric sensing (including enzymatic and electronic readers);
- (U//FOUO) spectroscopic techniques (integrated photonics); and
- (U//FOUO) spectrometric techniques (ion mobility spectrometry remains the most mature technology, but a lower false-alarm rate is desired for improved confidence).

(U) This is not an exhaustive list of all potential technologies; however, these technologies have achieved sufficient maturity and have demonstrated performance such that they are deemed the most probable solutions to address LIPCS operational uses. Based on the outcome of the survey, in the near-term, technological solutions that are most capable of outperforming the LIPCS while remaining within the size, weight, power, and unit cost constraints are impedance-based sensors and enzyme-catalyzed colorimetric chemistries. These technologies should be investigated in terms of vapor sensitivity and response time and for detection of persistent or low-volatility agents on surfaces.

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**(U) REMOTE CHEMICAL SENSOR DEVELOPMENT – CB10674:
CONCEPTS AND TECHNOLOGY ASSESSMENT
FOR LEAVE-IN-PLACE CHEMICAL SENSORS**

1. (U) INTRODUCTION

(U//FOUO) In accordance with the National Defense Strategy, the Joint Forces must preserve Warfighter lethality in contested environments through agile and adaptable modernization efforts.¹ Under the Chemical and Biological Defense Program's vision, the Joint Forces achieve this end in a chemical and biological (CB) contested environment through coordinated and integrated efforts that neutralize any CB threats that our adversaries may employ to gain an advantage in competition and standoff.² To preserve freedom of movement and action in the execution of small- and large-scale operations, the ground and littoral expeditionary forces can develop situational understanding to avoid CB threat agents by placing low size, weight, and power (SWaP); readily deployable; non-recoverable CB sensing capabilities in strategically important areas in the maneuver space. Such leave-in-place chemical sensor (LIPCS) devices, defined as miniature early-warning sensors for chemical hazards, should be deployable without drastically increasing Warfighters' physical and cognitive burden while leveraging existing capabilities where possible, (e.g., unattended ground sensors [UGS] employed by intelligence, surveillance, and reconnaissance [ISR] systems). Two LIPCS variants would be needed to address two distinct chemical threat circumstances: (1) early warning of the presence of non-persistent vapor-phase hazards, and (2) indication of the presence of persistent condensed-phase contamination on surfaces. The applicability of the LIPCS devices in these two circumstances is summarized in Table 1, along with appropriate performance characteristics as established by extant requirements.

Table 1. (U) LIPCS Variants and Performance Characteristics

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	LIPCS Vapor	LIPCS Solid/Liquid
Function	Give early warning of airborne hazards	Indicate the presence of persistent chemical hazards
How Deployed	In and around troop positions or friendly positions	In key terrain, named areas of interest, or along avenues of approach
Sensitivity	Immediately dangerous to life and health (IDLH)	Operationally hazardous aerial concentration levels
Size/Weight/Mission Duration (including batteries)	<1 in. ³ , ~1/3 lb, >8 h	

(U//FOUO) This report identifies several candidate concepts of employment (CONEMPS) for LIPCS devices. This understanding is critical to identify current technologies with potential for near-term assessment or advanced development, define relevant LIPCS performance parameters, and identify relevant near- and mid-term science and technology (S&T) concepts in this technology space. We engaged end users to identify conceptual use cases for this category of technologies, defined performance characteristics accordingly, and surveyed the technology space for suitable government and commercial off-the-shelf (GOTS and COTS, respectively) systems. In addition, we surveyed emerging technologies reported in the recent technical literature that show promise for addressing the performance characteristics of LIPCSs for unattended monitoring of chemical threats.

2. (U) SCENARIO DEVELOPMENT

(U//FOUO) The initial focus of this project was to develop a conceptual understanding of the operational space and scenarios for the deployment of LIPCSs. To accomplish this, we conducted interviews with users from the U.S. Army (USA), U.S. Navy (USN), U.S. Air Force (USAF), and U.S. Marine Corps (USMC), as well as other interested stakeholders over a six-month period. Through these discussions, the LIPCS concept and scope were defined. Individual Service input is presented first and is followed by a Joint Services summary that focuses on common themes that emerged during the discussions.

2.1 (U) Service Discussions

2.1.1 (U) U.S. Army

(U//FOUO) The USA combat developers appreciated the LIPCS concept and emphasized the application of the sensors for persistent monitoring of named areas of interest (NAIs) and avenues of approach (AAs; Figure 1). The value of the LIPCS to the USA in the execution of maneuver missions derives from the ubiquity of the sensors. Reconnaissance and surveillance (R&S) teams would emplace the sensors into NAIs prior to maneuver operations, and the sensors would be interrogated during the main, supporting, and reserve element movements. The USA combat developers suggested that a communications mode involving infrared (IR) diodes in lieu of radiofrequency (RF) transceivers could provide remote query and connectivity for the sensors. This would enable them to report their status to the approaching forces while alleviating the concerns surrounding the challenges of RF transmission from the LIPCSs. There was also interest in an unmanned aircraft system (UAS) or unmanned ground vehicle (UGV) that could remotely deploy the LIPCSs and later revisit their location and connect with them via the IR diode link to query their status (clean vs dirty), thus providing a georeferenced report on the presence of contamination along a route or in an NAI. As conceived, the LIPCS technology would provide satisfactory performance by developing a binary “threat/no threat” chemical hazard response (threshold) or class-based (objective) threat presumptive identification (blood, blister, or nerve). Ideally, the LIPCS sensors would respond to operationally relevant levels of persistent contamination on surfaces (as defined by current programs of record, including the Proximal Chemical Agent Detector and the Nuclear Biological Chemical Reconnaissance Vehicle Sensor Suite Upgrade [NBCRV SSU] for chemical surface detection). The same or

(U//FOUO)

complementary LIPCS devices should respond to airborne vapor- or gas-phase threat agents at or above immediately dangerous to life or health (IDLH) levels, augmenting the sensing function that is currently in the field and expanding the area covered to enhance early warning of an airborne threat regardless of changes in wind direction. The USA combat developers identified the notional LIPCS as a durable, expendable sensing capability with a series production price point of less than FY28 \$200 per item. The specific desired performance and SWaP characteristics included the capability to:

- (U//FOUO) detect a threat agent delivery event on key terrain;
- (U//FOUO) store a detection event in volatile memory until interrogated by a manned or unmanned (ground or air) vehicle;
- (U//FOUO) be employed by physical impingement into soft soil or sand via UAS, UGV, drop-off, munition delivery, or hand emplacement;
- (U//FOUO) be employed by airdrop from 500 ft above ground level;
- (U//FOUO) defeat radio direction-finding, jamming, and beaconing by use of a non-RF or no traditional RF-based telemetry solution;
- (U//FOUO) select mission duration;
- (U//FOUO) self-destruct at the end of the mission; and
- (U//FOUO) self-destruct post-employment, in the event of tampering.

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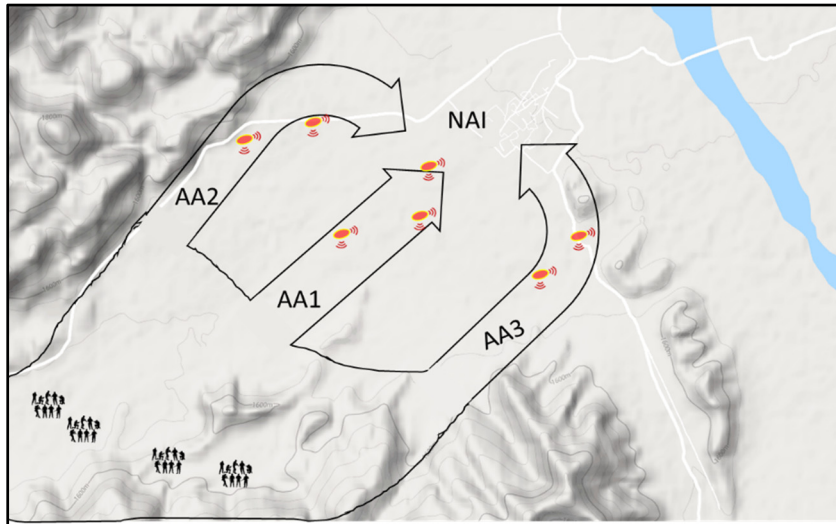


Figure 1. (U) LIPCS concept of use for deployment along AAs for persistent chemical presence monitoring.

(U//FOUO) USA user representatives expressed interest in the ability to camouflage the LIPCS so that they will not be readily discovered after emplacement. They were also interested in the ability to incorporate a self-destruct mechanism into the LIPCS that would engage upon evidence of tampering or by a remote command, which could be issued by manned and unmanned systems that approach the sensors to within sufficient range to establish communications with the devices. USA combat developers expressed a desire to integrate the LIPCS technology with existing non-CBRN UGS systems such as acoustic, passive IR motion detection, magnetic and seismic sensor arrays, and electro-optical imaging surveillance cameras. Such UGS arrays frequently include extended-life batteries augmented by solar panels for prolonged mission duration. Finally, the USA user representatives communicated a preference for the LIPCS sensor network to be integrated in such a manner that complementary remote sensing systems, including the Joint Services Lightweight Standoff Chemical Agent Detector or candidate refresh technologies, can be automatically cued to interrogate the airspace in the vicinity of an activated LIPCS and thereby alert to the presence of a suspected hazard and enhance early warning confidence and reliability.

2.1.2 (U) U.S. Marine Corps

(U//FOUO) The USMC combat developers envisioned dispersal of small, inexpensive, “disposable” sensors to detect chemical threat agents. For example, LIPCS could be strategically dispersed around rapidly configured listening or observational post (LP/OP) perimeters, to mitigate an inadequate number of Aerosol and Vapor Chemical Agent Detection (AVCAD) sensors. For this type of scenario (Figure 2), an audible alarm could be considered an alerting mechanism; however, there is concern about noise and light discipline under other mission sets. In that regard, multiple alarm modes and indicators are generally desired. Another application of interest is to monitor a route and augment other sensors, including the Tactical Remote Sensor System (TRSS). The TRSS is a UGS array for ISR applications that consists of electro-optical infrared cameras, thermal sensors, and ground surveillance radar kits that detect personnel and vehicles. There are light, medium, and heavy variants that comprise the Ground-Based Operational Surveillance System capability sets for deployment at different echelons. All data are monitored by a wired or wireless link and a cross-domain solution to the operator station at the Combat Operations Center. Current implementations leverage the Minotaur and Maven appliance for sensor data aggregation. The LIPCS could augment the TRSS capability sets for chemical hazard detection. The USMC end users offered that to provide operational benefit to Marine missions, the LIPCS technology must be very low burden, easy to deploy, and should deliver a high level of alarm confidence in the presence of airborne threat agents. The LIPCS would be expected to presumptively detect airborne vapor-phase threat agents in a binary threat/no-threat state (threshold) or by class of threat (blood, blister, or nerve) (objective).

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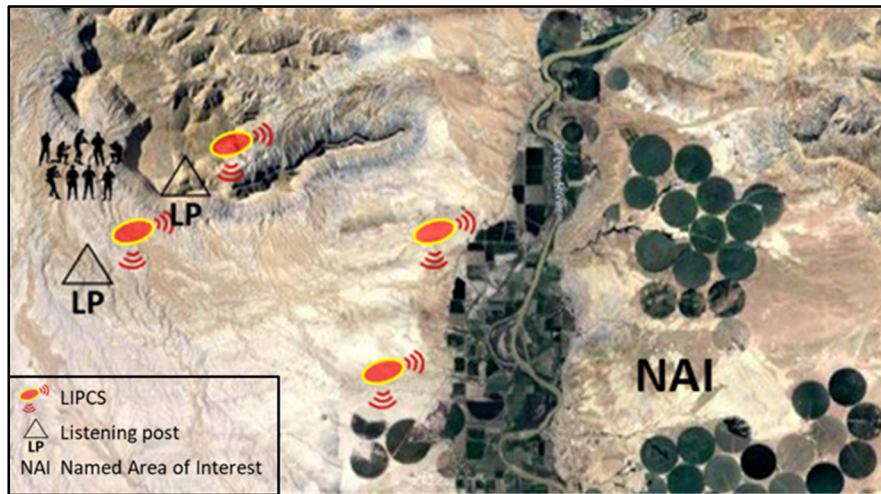


Figure 2. (U//FOUO) USMCS concept depicting LIPCS devices (orange circles) deployed near an LP/OP to augment early warning of threat agent vapors and at a distant NAI to monitor for adversary deployment of a persistent surface-contaminating threat agent.

(U//FOUO) For expeditionary air fields and forward operating base applications, the AVCAD is expected to be employed, but for expeditionary operations consisting of small teams with lighter equipment, toss-and-leave sensor products could be valuable. A CONEMPS that augments the capability delivered by and integrated with the Joint Chemical Agent Detector (JCAD) was also deemed desirable. In this application, LIPCS would detect airborne vapor-phase threat agents in a binary (threat/no-threat) state at levels that compare with JCAD capabilities.

(U//FOUO) The concept is that small teams could have one JCAD for the whole team, but each operator could carry several smaller LIPCSs and deploy them in tandem with the JCAD to provide better spatial coverage and leverage the performance of netted sensor arrays and algorithms (Figure 3). Operational parameters requiring further investigation include the maximum distance between sensors that will provide continuity in protection; the distance outside the perimeter required to provide enough warning; and the potential for a “Tom Brady wristwatch” interface to alert users to the state of the remotely deployed sensor. Such a capability is readily embodied in the USMC handheld computer, which is an Android platform comparable to the USA’s Nett Warrior handheld end user device. Any sensor deployed in this fashion would be placed far enough out that the alert could be transmitted to an unprotected team, which would then send an element out in mission-oriented protective posture to investigate. Further development of an operational view that illustrates this concept should be performed, particularly to help users envision how these might work to generate beneficial feedback and to show a scalable concept of employment that varies by team size.

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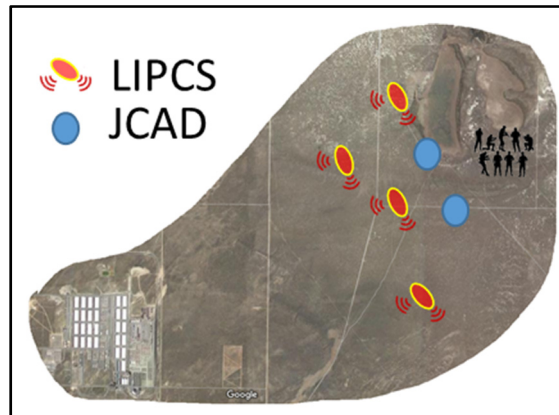


Figure 3. (U//FOUO) LIPCS devices deployed in tandem with JCADs to increase spatial coverage of hazard warning zone.

(U//FOUO) Another desired concept is an inexpensive “mini chem sensor” integrated on the USMC’s Nibbler UAS platform for the “quad for every squad” concept. The chemical detection module under investigation by CBRN Sensors Integrated on Robotic Platforms (C-SIRP) is an example. The Marine Corps Warfighter Laboratory is the lead for all concepts related to the Nibbler and develops Marine expeditionary force-level exercise opportunities for S&T insertion, providing opportunities to leverage S&T investments by multiple communities. The Marine Corps Intelligence Network and ISR capabilities like TRSS could serve as a hub for networked sensors under this concept (Figure 4).

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Figure 4. (U//FOUO) The USMC Nibbler drone is a 3D-printed capability controlled by the Marine Common Handheld computer, shown here with an LIPCS implementation payload.

(U//FOUO) As mentioned, the conceptual LIPCS appears to map to TRSS and other CONEMPS as an NAI monitor, or target area of interest (TAI), in which the sensors are pre-placed or even projected (dropped) into the site. Larger area facilities, main supply routes (MSRs), and a response to an expectation from intelligence authorities of an adversary action involving weapons of mass destruction (WMD) could also be monitored or protected with the capability. Light armored reconnaissance battalions, which are up to 7 km ahead of the main body, could be considered. For example, an intelligence mission could be leveraged so that sensors relay information over a communications relay drone. Deployment of LIPCSs at choke points and other canalizing terrain features along a route of advance would provide early notification of adversary action involving chemical threats and would also fall under these CONEMPS. The conceptual LIPCS employed in this fashion would respond to operationally relevant levels of persistent contamination on surfaces (as defined by current programs of record including Proximal Chemical Agent Detection and NBCRV SSU chemical surface detection). The same or complementary LIPCSs would respond to airborne vapor- or gas-phase threat agents at or above IDLH levels.

2.1.3 (U) U.S. Navy

(U//FOUO) In discussions with USN combat developers, the most logical mapping of the LIPCS would be in support of specialty teams whose missions involve entering unknown environments with potentially hazardous atmospheres, to include visit, board, search, and seizure (VBSS); explosive ordinance disposal (EOD); and sea, air, and land (SEAL) teams. The USN opted for components of the Dismounted Reconnaissance Sets, Kits, and Outfits acquisition program that include low-oxygen-environment sensors and low-exposure-level toxic industrial chemical or toxic industrial material (TIC/TIM) detection. The most likely conceptual use case of the LIPCSs by VBSS, EOD, SEAL teams, and Seabees would be in sensitive site exploitation missions, investigating potentially hazardous situations on an adversary's detained vessel, and on unmanned ground and air platforms. The LIPCSs might also be suitable for screening choke points ahead of a task force movement or at entry-control points in ports. One operational group discussed littoral operations as one of the most dangerous circumstances for potential adversary deployment of chemical hazards, and they suggested distributing LIPCSs on buoys or similar floating platforms around the beachhead and shoreline (Figure 5). To be effective in this application, the LIPCS should provide sensitivity and specificity sufficient to detect airborne vapor-phase threat agents in a binary (threat/no-threat) state at levels that compare with JCAD capabilities.

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Figure 5. (U//FOUO) SEAL teams deploying an LIPCS array to protect amphibious landing operations.

2.1.4 (U) U.S. Air Force

(U//FOUO) In discussions with USAF representatives and stakeholders, the LIPCSs could be components of an airfield protection and assessment CONEMPS, with the possibility of a line-powered configuration (Figure 6). The USAF notes that the sensors would be emplaced in locations where they are vulnerable to damage or loss by kinetic events on the airfield, so it is important that they be low-cost and supported by a logistics chain of replacement items. The USAF would be interested in an LIPCS to streamline or automate current airfield protection, for vapor sensing and liquid contamination sensing. Teams of airmen must physically visit each position on the airfield to inspect the sensors and papers for the presence of vapor or persistent surface contamination. The USAF also desires that LIPCSs additionally respond to aerosols, low-volatility agents, mid-spectrum agents, and pharmaceutical-based agents. An electronic interface that conveys the state of the LIPCS for both vapor hazards and surface contamination would reduce or eliminate the need for airmen to physically inspect protected areas. A USAF initiative called Rapid Airfield Damage Assessment System (RADAS) comprises a host of technology packages for inspection of airfield surfaces and facilities after a kinetic event, to support rapid quantification of any residual damage that could affect airfield serviceability for launching and landing sorties. The LIPCSs could augment the RADAS to enable sensing of chemical contamination. To satisfy USAF expectations in such applications, LIPCSs would respond to operationally relevant levels of persistent contamination on surfaces (as defined by current programs of record, including Proximal Chemical Agent Detection and NBCRV SSU chemical surface detection). The same or complementary LIPCSs would also respond to airborne vapor- or gas-phase threat agents at or above IDLH levels.

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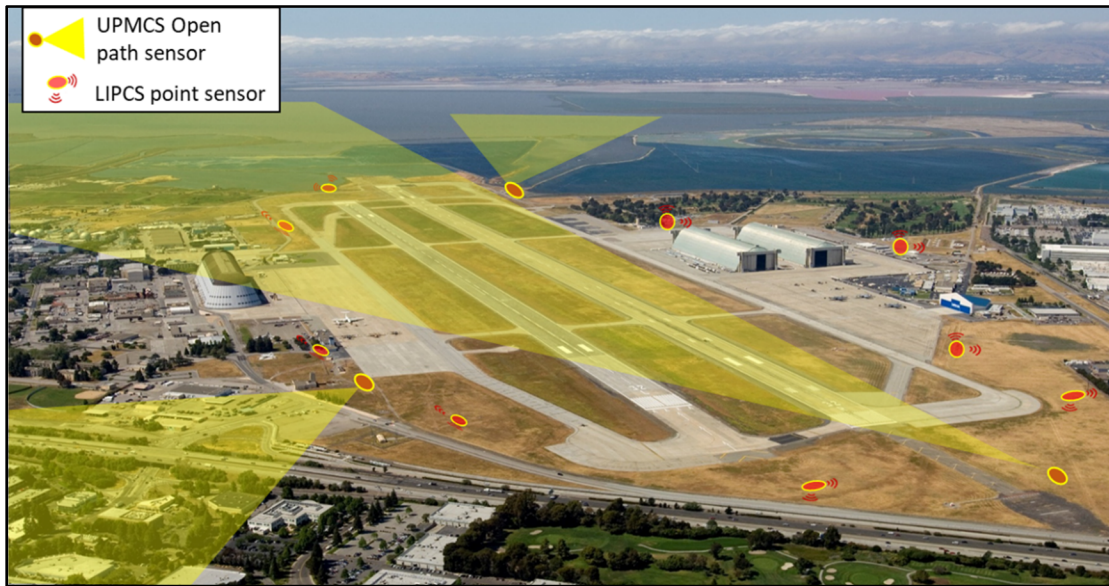


Figure 6. (U) LIPCS sensors (orange circles) deployed along with unattended perimeter monitor chemical sensors (yellow triangles) to monitor airfield for contamination.

2.1.5 (U) U.S. Special Operations Command

(U//FOUO) USSOCOM concept developers assessed that the LIPCS would have utility when deployed along MSRs or AAs. LIPCSs would also be valuable when deployed forward of the forward line of troops; for example, by a reconnaissance force emplaced in support of the maneuver plan (AA or direction of travel). Unattended sensors may also have value when deployed along phase lines, sector boundaries, or coordination points. They could be emplaced to rear or lateral troop locations to support off-target attacks when the wind direction is not from the front of the formation. Another concept of interest to U.S. special operators would be a layered defense array that fuses information to provide a “standoff integrated early warning”. Special operators would support intra-theater mobility assets by providing area monitoring and alert-in-depth via unattended LIPCS devices deployed by manned rotary-wing aircraft and UASs, mortar tubes, grenade launchers, or artillery. To satisfy USSOCOM performance expectations in such applications, the LIPCSs would respond to operationally relevant levels of persistent contamination on surfaces (as defined by current programs of record including Proximal Chemical Agent Detection and NBCRV SSU chemical surface detection). The same or complementary LIPCSs would respond to airborne vapor- or gas-phase threat agents at or above IDLH levels.

(U//FOUO) USSOCOM also illustrated several special operations forces applications for the LIPCSs, including the following:

- (U//FOUO) overt, covert, clandestine, and low-visibility emplacement of LIPCSs at or on the objective area to support reconnaissance, characterization, and post-attack damage assessments;

- (U//FOUO) LIPCSs emplaced on selected weapons at WMD sites or CBRN hazards to monitor and support contamination avoidance, force protection, increased situational awareness, and post-event hazard mitigation and control;
- (U//FOUO) LIPCSs or unattended sensors employed over an extended operating range, such as roughly 100 m or more (versus the 10–20 m baseline achievable using a JCAD or other organic sensor) and supported through UAS- or ground-emplaced retransmission stations; and
- (U//FOUO) LIPCSs or unattended sensors embedded into form factors designed to blend into multiple operating environments (i.e., urban, woodland, desert, facility, etc.) with a low observable form factor (e.g., a native rock, brick, building vent, light fixture, etc.).

2.2 (U) Chemical Biological Operational Analysis

(U//FOUO) In addition to our discussions with individual Services, we presented the LIPCS concept to end user participants at the Chemical Biological Operational Analysis (CBOA-2019; held at Camp Dawson, WV) as part of the “concept tent” to gather feedback on the technical concept before launching any significant S&T efforts to develop the technology. At CBOA-2019, we had discussions with members from all Services and other DoD agencies, usually after operators returned from the daily field trails and exercises in which CBRN scenarios were carried out. This allowed us to directly obtain feedback on how LIPCSs could have been employed in that day’s exercise to mitigate unforeseen gaps or needs. The LIPCSs were described to the participants as small, inexpensive, “disposable” sensors that could be scattered around to indicate the presence of chemical threats. We stressed that the intent of the LIPCSs was that Warfighters could carry several of them and toss them into place during missions or in critical nodes like NAIs (Figure 7), or they could drop them like breadcrumbs along a route of advance during an early entry or the route reconnaissance phase of the operation (Figure 8).

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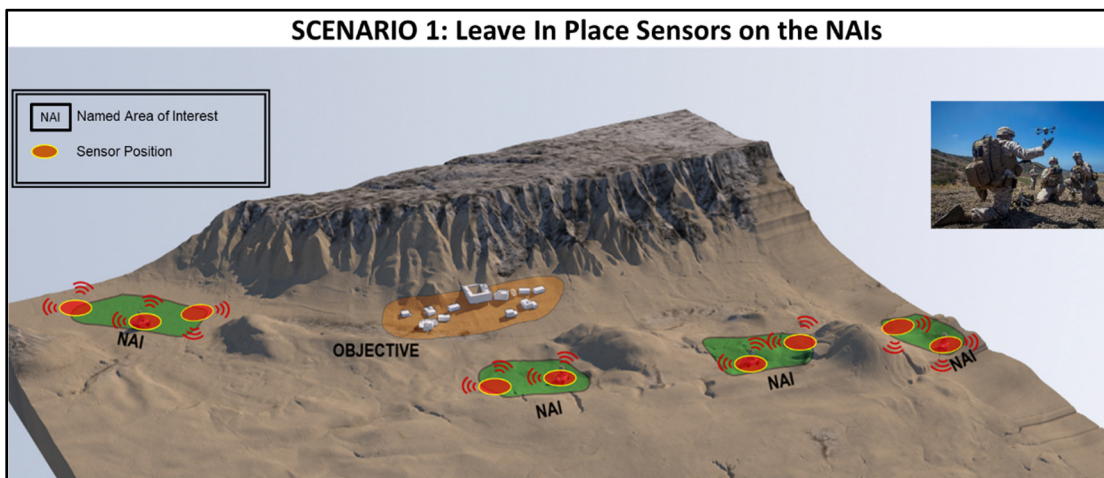


Figure 7. (U//FOUO) Scenario 1: LIPCSs on the NAIs. Sensors are deployed by reconnaissance element or projected into NAIs. Sensors are periodically interrogated (e.g., by a small UAS that pairs with the sensors to read out their detection status and indicate threats). These could also be synchronized with other ISR systems emplaced in the NAIs.

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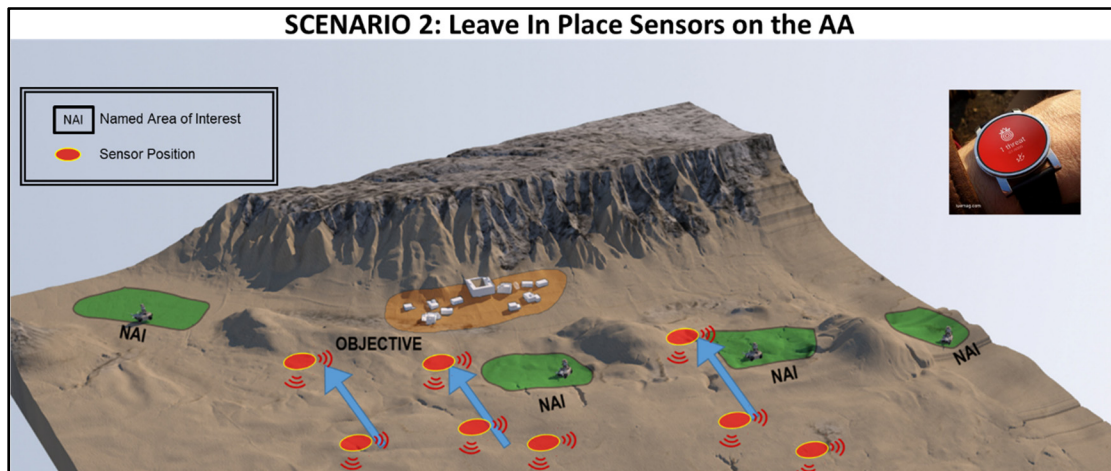


Figure 8. (U//FOUO) Scenario 2: LIPCSs on the AA. Sensors are deployed along an AA by reconnaissance teams or are projected ahead of a maneuver mission. On approach, sensors could be interrogated by a handheld device that pairs with the sensors to read out their sensing outcome. These could also be synchronized with other ISR systems emplaced on the battlefield.

(U//FOUO) We confined discussions on LIPCSs to chemical detection while noting that there is a parallel program concept being explored for biodetection. For example, a BioACER (Biological Automated Collector/Detector for Expeditionary Reconnaissance) prototype was also on display in the concept tent. This system offers a compact, deployable capability that implements loop-mediated, isothermal amplification, polymerase chain reaction assays in a form factor that is compatible with air-dropped arrays. LIPCS discussions were focused on the need to develop technologies that lower operator burden while supporting operation for extended periods of time and providing personal immediate warning of a conventional chemical agent or emerging threat agent. The technology would also benefit from implementation of critical infrastructure-protection systems in which the sensors could operate on shore power for indefinite periods of time. Sensors that are deployed “outside the wire” should blend in or be camouflaged to look like indigenous rocks or other natural objects, and they should possibly self-destruct when tampered with or when nearing the end of their mission life.

(U//FOUO) The LIPCS concept resonated with small unit operations because it affords the capability to drop sensors from a payload on a small UAS or to carry and toss the sensors when concerns exist about possible chemical agent use by an adversary. Key functionality should include operation under various terrain-specific circumstances at sufficient distance from the operative teams to provide a detect-to-warn capability. The specific distances depend on mission, environment, troops, terrain, and time. The systems would be most useful if integrated with existing user interfaces, such as the Nett Warrior end user device (EUD) or the USMC handheld computer (both are Android-based computational environments), in lieu of adding another controller or interface.

(U//FOUO) The LIPCS concept could provide benefits for Title 10 and Title 32 Defense Support to Civil Authorities missions, such as reconnaissance of various confined spaces or pockets that can trap hazardous vapors and aerosol threat agents after an incident. It would also

(U//FOUO)

be valuable for inspecting confined spaces and interrogating areas where secondary explosion hazards may be present. The sensors should be droppable into confined spaces, be able to establish communications with a controller hub, and capable of functioning on battery power for at least 8 h.

(U//FOUO) Several operators commented that the LIPCSs should be simple and lightweight. We interpreted that to mean that the sensor package should be small enough that three to five individual sensors complete with all electronics for communications and remote LIPCS operation can easily fit into the space of a single rifle magazine or cargo pocket. The sensors should have the capacity to alert Warfighters either audibly or visually in force protection applications, or silently via remote telemetry to a handheld computer or Nett Warrior EUD, in R&S missions in which noise and light discipline is paramount. As in previous discussions with individual Services at CBOA-2019, operators wanted to drop the devices well in advance of a maneuver operation for route reconnaissance and in NAIs, perhaps interrogating them via UAS or along the approach route by the forward elements of the maneuver force.

(U//FOUO) As part of the CBOA-2019 event, we submitted a questionnaire that participants were asked to fill out following our discussions on LIPCSs. Over the three-day event, we interviewed and interacted with more than 70 users from across all of the Services, as well as other DoD-related stakeholders. The instantaneous feedback we received was invaluable and is included herein. However, less than 15% of the participants we spoke with completed and submitted the questionnaire. From the returned questionnaires, we obtained feedback from infantry and CBRN specialists, comprising enlisted personnel and officers from the USA and USMC, with a range of experience from 1 to 26 years. The feedback from the questionnaires was overwhelmingly positive toward the LIPCS concept. Experienced Warfighters (with 3+ years of service) highlighted that this technology could close capability gaps in their respective mission sets. Table 2 summarizes the responses.

Table 2. (U) Condensed Summary of Responses from CBOA Questionnaire
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Question/Area of Interest	Response
Do remote chemical sensors fill mission shortfalls?	Yes: enhanced detection and situational awareness
What is the probable mission impact?	Reconnaissance/detect to warn
Is the logistical burden acceptable?	Yes: additional trainings and use case understanding needed
What additional functions are desired?	Detection of other threats, aquatic use

3. (U) LIPCS CONCEPT, SCENARIO, AND REQUIREMENTS

(U//FOUO) Through the individual Service discussions and CBOA-2019 event, several scenario-based concepts of use for LIPCS emerged, as well as a standard request for needed requirements.

3.1 (U) Conceptual Scenarios for LIPCSs

(U//FOUO) A clearly portrayed concept is key for operational understanding and CONEMPS building for the users of LIPCS. A common functionality use scenario mentioned in each of the Service discussions (Section 2.1) was for LIPCS sensors to be used for R&S purposes, albeit in different situations. The following list is not a comprehensive, but it captures the general use case scenarios:

- (U//FOUO) deployment of sensors around NAIs and TAIs for R&S (outside-the-wire placement);
- (U//FOUO) deployment of sensors around airfields, logistics support areas, and ports to continuously monitor for vapor hazards and surface contamination;
- (U//FOUO) deployment in choke points for route surveillance, clearance, and reconnaissance and in egress and ingress points (to ensure exit point is contaminant free);
- (U//FOUO) deployment of sensors for subterranean room and hallway reconnaissance (ability to roll or lob a sensor into an area without physically breaching the area of interest); and
- (U//FOUO) deployment of sensor configuration to establish a safe area around structurally unstable or collapsed infrastructure.

3.2 (U) Proposed Requirements for LIPCSs

(U//FOUO) During discussions about the potential requirements of LIPCSs, users stated, *"It is less important to identify the threat than to classify the presence of a threat."* In general, the LIPCS is expected to be a less specific and potentially less sensitive sensing and alerting capability than the standard JCAD. It must indicate anomalous circumstances consistent with a threat scenario and perhaps broadly classify the situation as a possible chemical attack. That said, the systems should still be deployed as a detect-to-warn-based functionality.

(U//FOUO) The network part of the technology is considered critical. There are logistics, calibration requirements, and other concerns. Particularly in the case of the LIPCS, the logistical and maintenance burdens should be kept to a minimum:

- (U//FOUO) Sensors should be approximately 1 lb or less, and optimally, ~1/3 lb (single round ammunition up to cartridge size).
- (U//FOUO) Sensor distance to operator should suffice for detect-to-warn (mask-up) capability.

- (U//FOUO) Different modes of alarm operation are needed (silent, transmits to LP/OP), audible, or visible (read by external source).
- (U//FOUO) Different modules should be operable by specialists and generalists.
- (U//FOUO) Battery life must be 8–12 h (threshold); 24 h or more (objective) is desired.
- (U//FOUO) Anomaly detection (threshold) and classification of threat (objective) capabilities are needed.
- (U//FOUO) Connectivity should be technology agnostic: connectivity to Nett Warrior, Marine Corps handheld, and other systems should be possible.

4. (U) TECHNOLOGIES UNDER CONSIDERATION

(U//FOUO) Following discussions with users to define initial scenario and technical requirements for LIPCS, we reviewed current commercial and non-commercial technologies, as well as known S&T programs that could address LIPCS development. The goal was to identify near-term system design concepts or areas of research and identify technological development opportunities. As stated earlier, the main hazard state that is being targeted by this study is vapor chemical threats. The secondary focus is persistent liquid and solid threats deposited onto surfaces.

(U) Multiple internal meetings were held at the U.S. Army Combat Capabilities Development Command Chemical Biological Center (CCDC CBC; Aberdeen Proving Ground, MD). Scientists and engineers from across the center discussed the user feedback, the S&T efforts, as well as potential solutions that could address the LIPCS challenges. A broad list of technologies was discussed, including the following:

- (U) electrical impedance-based techniques such as graphene, metal–organic frameworks (MOFs), molecular imprinted polymers (MIPs), and functionalized carbon nanotubes (CNTs);
- (U) spectrometric techniques (ion mobility spectrometry [IMS]) and print-on-demand IMS (tailor sensor for mission);
- (U) colorimetric sensing, including enzymatic and electronic readers; and
- (U) spectroscopic techniques, such as integrated photonics.

(U) These technologies are discussed herein, and potential LIPCS capability solutions are addressed.

4.1 (U) Electrical Impedance

(U) For this report, the term impedance is used to capture a wide range of technologies that incorporate various aspects of the impedance phenomenology and can include capacitance measurements, chemiresistors, electronics-based colorimetrics, and others.

(U) Electrical impedance (Z) is broadly described as the measurement of the dielectric constant generated by an alternating current as it flows through a circuit made of resistors,

(U)

capacitors, inductors, or any combination of these. For chemical detection purposes, the resistance of a dielectric medium may change due adsorption or interaction with various species. Device development centers on the detection of these changes, which are prone to environmental perturbations. Capturing the target analyte usually requires a surface functionalization, such as graphene complexes, MOFs, MIPs, functionalized CNTs, self-indicating colorimetrics (SICs), and other materials, including antibodies and aptamers when applied to biological detection.³⁻⁵

(U)

The use of materials that target hazardous analytes and exhibit changes in the dielectric constant upon exposure is an area that has not been well explored. Because the dielectric constant of a material can be tuned to change, based on specific interactions, this approach offers potential specificity for point detectors geared toward vapor sensing.

(U)

Impedance spectroscopy can also identify defects; for example, grain boundaries, trap states, and poor connectivity with contact leads. Such defects or contact-grain interactions may result in resistive changes that oppose that of the sorbent material upon exposure, or irreversible changes due to a secondary exposure (O_2 , relative humidity, salty atmosphere, smoke emissions, etc.). In such a case, impedance spectroscopy can identify these mechanisms, and the corresponding adjustments to the sensing material or device fabrication methods can be optimized. Common approaches that can be employed for impedance measurements are (1) the use of an interdigitated electrode (IDE), wherein a dielectric material can be coated onto the “finger region” of a circuit, and (2) the dielectric material is placed between two parallel plates that measure changes to the reactive capacitance due to sorption of chemicals across the media. Changes in the dielectric constant due to sorption of vapors, aerosols, solids, and liquids result in charge transfer to the electrode. Figure 9 shows both types of device-like structures.

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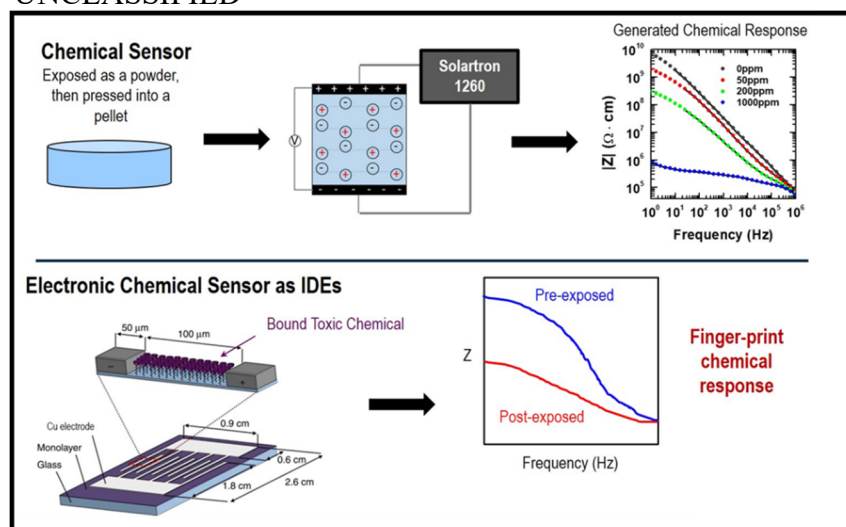


Figure 9. (U) Top: dielectric material (light blue) sandwiched between electrode plates. Bottom: film of the dielectric material deposited on finger region of IDEs.

(U) Advantages of integrating dielectrics into sensors:

- (U) simplicity of device can make it easily useable by generalists and specialists;
- (U) passive sampling device (i.e., no need for external power, pumps, etc.);
- (U) portability: extremely small size, weight, and power;
- (U) low costs upon mass production;
- (U) tunability of sorbent or surface-responsive materials for wide range of targets;
- (U) rapid response times (<10 s); and
- (U) very low SWaP.

(U) Weaknesses associated with integrating dielectrics into sensors:

- (U) accounting for atmospheric conditions (can cause unwanted responses);
- (U) sensitivity in observing low signal changes among the background; and
- (U) specificity: can be addressed with structural functionalization; however, each functionalization will have unique challenges in reproducibly coating or depositing sorbent on impedance-based devices.

(U) Modeling the impedance data can also provide a facile route to deconvoluting the data, that is, making it easier to identify each analyte species among environmental factors (pulling signal out of background). This technique has been explored for chemical warfare agents (CWAs) and explosives and in biological sensing with various levels of success.⁶⁻⁸

(U) The remainder of this section discusses current research and development of impedance-based sensors, as well as commercial and industry applications.

(U) Figure 10 shows two recent examples that demonstrate the broad applicability of electronic and impedance-based sensors and show their miniaturization potential. Figure 10A illustrates preliminary vapor-based detection at CCDC CBC with functionalized interdigitated capacitors (IDCs) coated with MOFs that selectively absorb complex nitrate-based explosives. Figure 10B is a more recent concept, showing a fully integrated, impedance-based sensor functionalized with SIC materials for detecting organophosphate-based chemicals. This sensor was recently developed at CCDC CBC and patented for CWA detection.⁹

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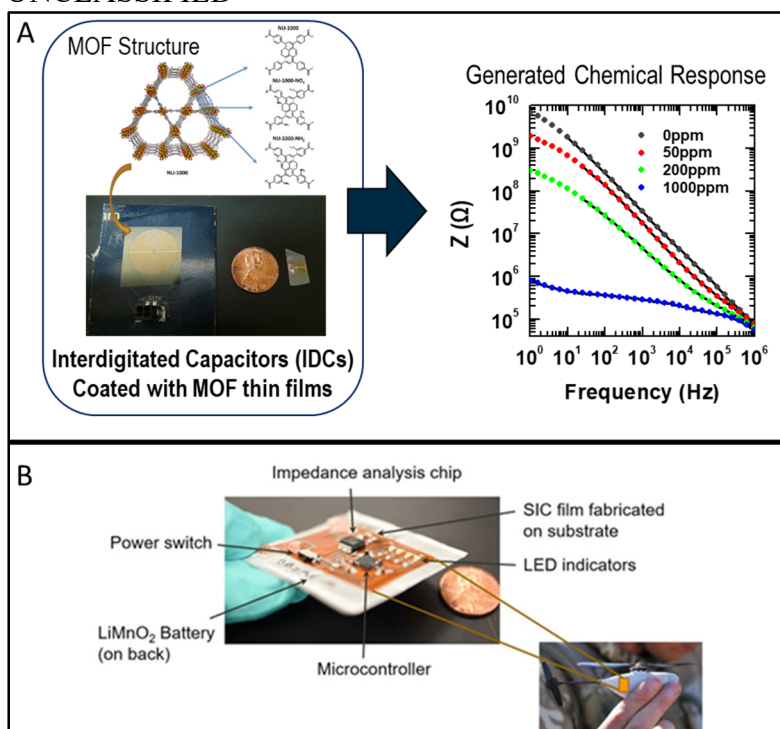


Figure 10. (U) Impedance-based chemical sensor showing small SWaP. (A) Functionalized IDCs coated with MOFs selectively absorb complex nitrate-based explosives. (B) Impedance-based sensor functionalized with SIC materials for detecting organophosphate-based chemicals.

(U) Commercial sensor prototypes are being developed that implement surface-functionalized impedance transduction to effect environmental sensing. Design West Technologies (Tustin, CA) develops functionalized CNT sensors aimed at integrated microsensor applications for various TIC vapors. Under a Small Business Innovation Research (SBIR) project for the Department of Homeland Security (DHS; Washington, DC), the firm developed and tested a microsensor against a host of TIC vapors, including ammonia, arsine, hydrogen cyanide, hydrogen sulfide, sulfur dioxide, and nitric oxide, with the aim of developing a small “sensing badge” to indicate the presence of these compounds at or above the permissible exposure limits established by the Occupational Safety and Health Administration (Washington, DC). More recently, they developed a handheld ammonia detector for DHS Countering Weapons of Mass Destruction Office, and are exploring the applicability of the technology for chemical agent sensing under a SOFWERX (Tampa, FL) initiative (Figure 11). Initial tests of the technology against sarin vapors yielded low performance in terms of sensitivity, and the company is investigating the possibility of incorporating a pre-concentrator.

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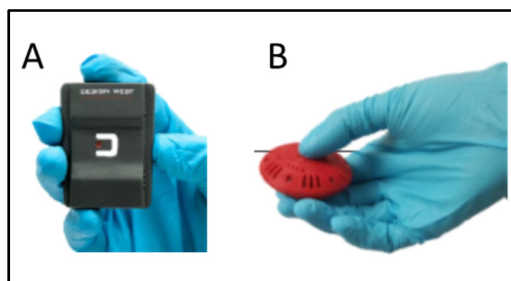


Figure 11. (U) Form factor of the Design West Technologies CNT microsensors: (A) handheld or badge module; (B) throwable puck module.

(U) Vaporsens (Salt Lake City, UT; vaporsens.com) developed an integrated sensor array consisting of multiple CNT impedance-based sensing elements, each of which was functionalized differently to provide a multivariate “signature.”¹⁰ The Vaporsens technology was recently evaluated under a series of field trials sponsored by the Defense Advanced Research Projects Agency (DARPA; Arlington, VA) SIGMA+ program. The firm developed 16- and 32-element arrays under this effort and deployed both variants (Figure 12). The test events revealed a significant and inexplicable level of response variability on the part of the sensor arrays, some of which the firm attributed to variations in temperature, humidity, and other environmental effects. The firm also engineered an air-sampling flow cell to optimize the performance of the sensor array. Finally, the sensing elements themselves exhibited hysteresis in their response to targeted compounds; after exposure, the impedance response only very slowly, if ever, returned to baseline.

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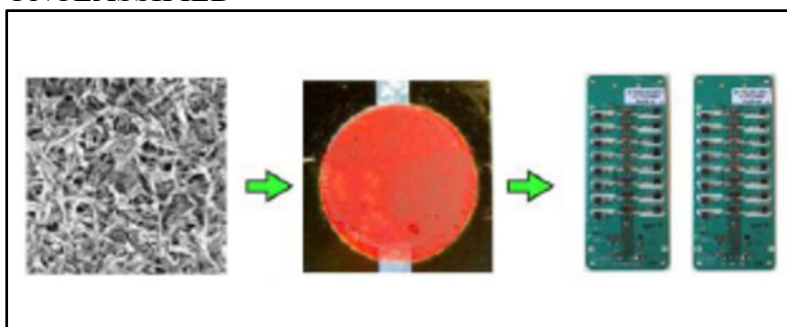


Figure 12. (U) Vaporsens CNT array technology. Carbon nanofiber sensory materials, each uniquely functionalized through covalently bonded dopants (left) were coated onto interdigitated electrodes (center) on a circuit board that housed 16 sensing elements. A complementary pair of 16-element sensing arrays (right) provides 32 uniquely functionalized sensing surfaces.

(U) A few other groups and firms developed products based on chemiresistive sensing arrays, including the National Aeronautics and Space Administration Ames Research Center (Mountain View, CA) and GE Global Research (Niskayuna, NY). The latter advanced their multivariate sensing concept to a networked sensor arrangement to implement the chemiresistive

(U)

sensing medium into resonant circuits on radiofrequency identification (RFID) tags, derive capacitance and resonant impedance from the arrays, and process the response data using machine learning techniques.

(U)

The Defense Threat Reduction Agency (DTRA; Fort Belvoir, VA)-sponsored Dynamic Multifunctional Materials for a Second Skin program incorporated advanced research by the Swager group at Massachusetts Institute of Technology (MIT; Cambridge, MA) to develop microsensor technologies based on functionalized CNT, graphene, and graphite; conducting layers modified by metal complexes; conductive polymers; metal oxides; or inorganic semiconductors to fabricate impedance-based sensing modalities.¹¹ The functionalized CNT sensor technology achieved a high level of technological maturity, including integration into commercial RFID tags passively powered and remotely interrogated by an RFID-enabled device (e.g., a smartphone; Figure 13).¹²

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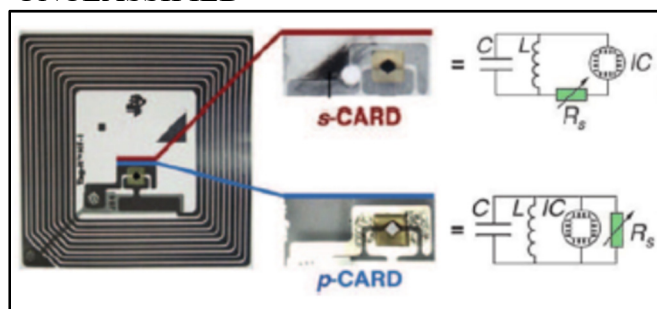


Figure 13. (U) Left: two methods to integrate CNT chemiresistors into commercial RFID tags. Right: the equivalent circuit diagrams.¹³

(U)

The Swager group successfully tested the CNT sensor technology as applied to trace detection of toxic chemicals¹⁴ and demonstrated their implementation with smartphone interrogation using these passively powered RFID tags.¹⁵ In addition, several studies aimed specifically at the detection of CWAs using functionalized CNTs by either taking the multi-element array approach¹⁶ or attaching various chemically selective compounds to the CNT surfaces. Notable compounds that exhibit some selectivity for nerve agents include acetylcholinesterase,¹⁷ hexafluoroisopropanol (HFIP),^{18–20} HFIP-substituted polythiophene,²¹ poly(3,4-ethylenedioxythiophene),²² tetrafluorohydroquinone,²³ and even single-stranded DNA.²⁴ The surfaces of CNTs electrochemically modified with ferrocene–amino acid conjugates exhibit exceptional sensitivity to nerve and blister agent surrogates.^{25–27}

(U)

Graphene is being advanced as a chemical sensor transduction material. Noteworthy work has been done by Polestar Technologies (Needham Heights, MA) to apply MIPs that were specifically engineered for CWA detection under their previous SBIR program (contract FA8650-14-M-5078). The firm developed sensors for detection of nerve agents soman and tabun by templating the binding layer using nonhazardous organophosphates (diethylchlorophosphonate and diethylcyanophosphonate, respectively) that have similar

(U)

chemical structures. Sensor performance was optimized by binding layer monomer and dimer systems that (1) exhibit strong molecular interactions with chemical substituents of the target molecules selected using molecular dynamics modeling, (2) employ an optimal ligand–receptor stoichiometry, and (3) incorporate cross-linking to reduce thermally induced binding site deformation.²⁸ The MIP-functionalized graphene sheet exhibits a clean, readily sensed RF backscatter spectrum sensitive to binding at the polymer’s MIP-functionalized receptor sites when interrogated by a remote RF analyzer that includes the graphene–MIP assembly integrated onto an RFID tag (Figure 14).²⁸ Figure 15 shows the repeatable response and marginal sensor-to-sensor variability of replicate sensor assemblies exposed to 0.2 and 2 ppb soman and tabun.

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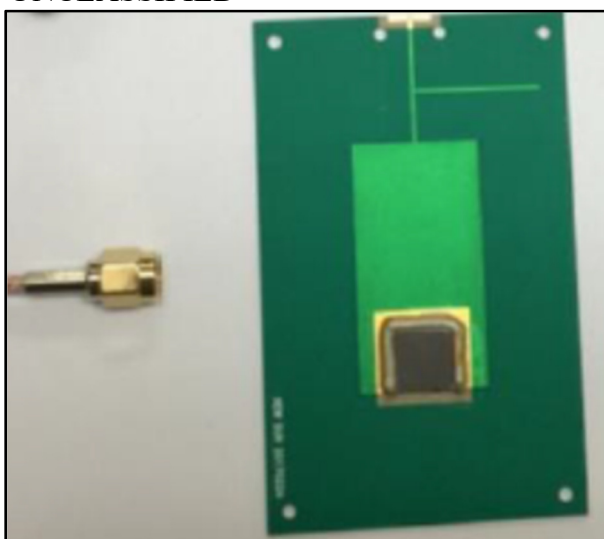


Figure 14. (U) Polestar Technologies MIP-functionalized graphene sensor RFID assembly. A standard BNC (Bayonet Neill-Concelman) cable connector is shown for scale.

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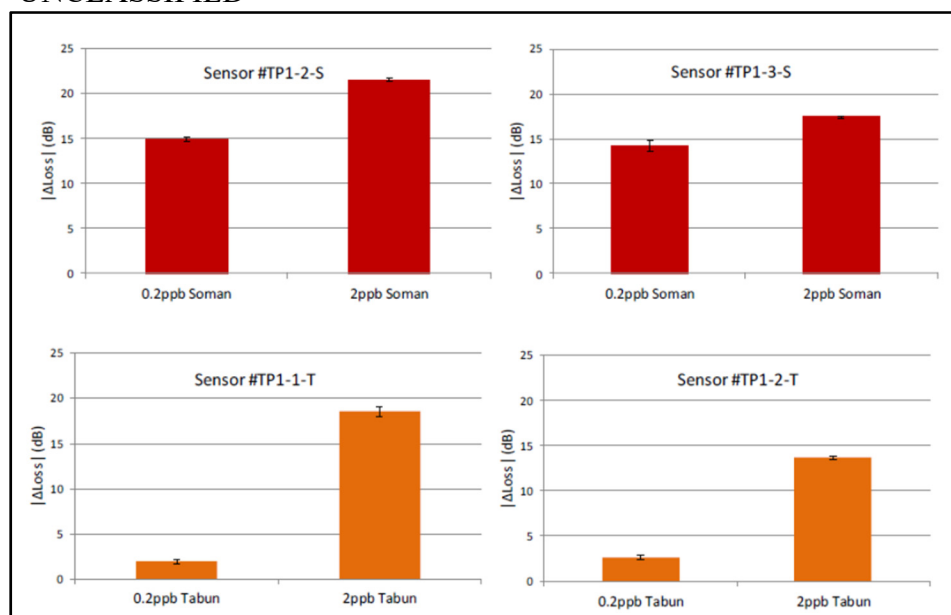


Figure 15. (U) Responses of MIP-functionalized graphene sensor RFID tags upon exposure to 0.2 and 2 ppb levels of soman (top) and tabun (bottom).

(U) The Polestar Technologies MIP-functionalized graphene sensors did exhibit some cross-reactivity to humid conditions (at 45% relative humidity) and to fuel vapors (JP-8 at 1% of its saturation level). The high-humidity results also exhibited a possible interaction with the higher concentrations of CWA vapors, as evidenced by an apparent reduction in the response to these vapors under humid conditions.

(U) Graphene exemplifies two-dimensional (2D) nanomaterials whose macroscopic characteristics lend themselves to sensing applications by virtue of their extraordinary chemical and physical properties. In addition to graphene nanostructured films, which have the unique property of being only one atom thick yet upwards of several square centimeters in spatial extent, additional synthetic materials including phosphorene, hexagonal boron nitride, transition metal dichalcogenides, layered metal oxides, 2D MOFs, covalent organic frameworks, and other 2D compounds.²⁹ The common theme in the transduction mechanism is the exceptionally high surface-area-to-volume ratio of the 2D materials, which gives rise to exquisite sensitivity to binding events, in the form of electrical responses such as current, capacitance, resistance and impedance, voltage, and electrical potential.³⁰ Several firms have integrated impedance-based sensing transducers that are based on metal oxide semiconductor assemblies, including the UltraKera sensor (Integrated Device Technology; San Jose, CA), the Versatile Wearable Gas Sensing Platform, and the Voltor sensor (Ashwin-Ushas Corporation; Marlboro, NJ). The performance of this family of sensors relies on both the 2D material and its functionalization. Ashwin-Ushas Corporation developed a plethora of unique, one-time-use derivatized electrode-functionalized surface layers that interact with targets and respond voltametrically. Although this promises a higher degree of specificity, the readout electronics are more sophisticated, complicated, and expensive than those required for a simple resistance measurement. Another promising technology that departs from the standard chemiresistive transduction mechanism is

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the SC-210 micro-electro-mechanical system chemicapacitive sensor array (Figure 16; Seacoast Science; Carlsbad, CA; <https://www.seacoastscience.com>). The underlying signal transduction mechanism of the SC-210 system is proprietary. It is described as a modular sensor capability with an interchangeable sensor array that allows the system to be customized for specific missions or target analytes. The sensor weighs under 100 g and can be interfaced with the Seacoast Science miniature gas chromatograph as well as other modules that tune the system's performance. Unfortunately, few details regarding the performance of this technology are available.

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Figure 16. (U) Seacoast Science SC-210 modular chemical detection system.

(U)

Strle and coworkers investigated silane-functionalized, alternating-current-coupled chemicapacitive sensor arrays in electronic nose applications. They demonstrated the rich chemometric signature space generated by these constructs and the use of deep learning training methods to develop reliable chemical detection and recognition algorithms. The group constructed differential capacitor pairs tuned to minimize the difference in capacitance between the microcapacitors in each pair, thereby allowing the sensing system to be operated at high gain for optimal sensitivity. Their approach achieved remarkable selectivity for energetic compounds, with a sensitivity of 3.5 ppt per root hertz.³¹ Dr. Rod Kunz suggested that an integrated circuit (IC) foundry could make the differential chemi-capacitance chips, and Seacoast Science could functionalize and integrate them into low-cost, high-performance electronic sensors.³² A host of emerging advanced machine learning and signal processing approaches have demonstrably improved the selectivity and sensitivity of electronic nose constructs.³³

(U)

A team at the U.S. Army Combat Capabilities Development Command Soldier Center (CCDC Soldier Center; Natick, MA) is developing a chemiresistive vapor sensor using a random polymer-graphene nanoplatelet composite.³⁴ The composite exhibits electrical properties similar to those of the graphene sheets, but it is less expensive and easier to make. The team is investigating the performance of a 24-element sensor array in which each graphene sensor is uniquely functionalized for responsivity to different trace vapors (Figure 17). The team is also investigating the functionalization of their sensors with antimicrobial peptides to serve as a biothreat agent detection modality.

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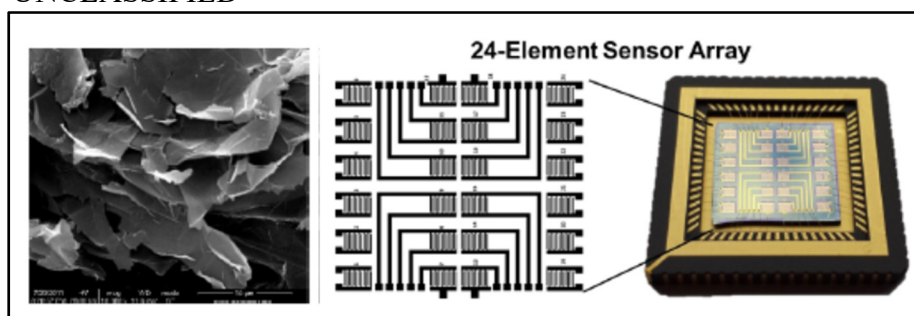


Figure 17. (U) An assortment of uniquely functionalized polymer-graphene nanoplatelets (left) coat each of the 24 sensor elements on an array construct (right) in a prototype electronic nose sensor concept developed at CCDC Soldier Center.

4.2 (U) Spectrometric Techniques (IMS)

(U) IMS is one of the most widely used and mature technologies for chemical vapor sensors and is the underlying technology in the currently fielded JCAD systems.

(U) Ions are generated by a corona discharge source at the inlet of the spectrometer and pulsed into a drift tube under the influence of an electric field. The ion mobility spectrum develops when ions move against a moderate backflow of air and impact a series of collector plates along the length of the drift tube. The spectrum comprises the reduced ion mobility (K_0) value of each spectrum peak. The K_0 value is calculated by normalizing K against standard temperature and pressure:

$$(U) \quad K_0 = K \left(\frac{273.15}{T} \right) \left(\frac{P}{760} \right) = \frac{L^2}{V t_d} \left(\frac{273.15}{T} \right) \left(\frac{P}{760} \right) \quad (1)$$

(U) where T is the drift gas temperature (in K), and P is the drift gas pressure (in Torr). This description of IMS applies to drift tube IMS, which is the separation of ions and measurements of K_0 values under a constant weak electric field. Many other forms of IMS exist, including, but not limited to, traveling wave IMS, differential or field-asymmetric IMS, and trapped IMS. Some IMS-based technologies do not use the shutter grid as a part of the ion separation technique. However, all IMS-based technologies perform velocity measurements of the reactant and product ions moving under the influence of electric fields: all IMS-based instruments contain an ionization source, a reaction region, an ion drift (separation) region, and an ion collector.³⁵ In general, IMS sensors have the following advantages and disadvantages.

(U) Advantages of IMS sensors:

- (U) simplicity of device can make it easily useable by generalists and specialists;
- (U) sensitive (parts per billion range);
- (U) rapid response times (≤ 2 s); and
- (U) pseudo quantitative.

(U) Weaknesses associated with IMS sensors:

- (U) consumables and their disposal (molecular sieve pack);
- (U) clear-down times (can be long, depending on chemical sorption and desorption);
- (U) presumptive detection;
- (U) subject to false alarms; and
- (U) larger SWaP than desired for LIPCS.

(U//FOUO) Areas of current and potential IMS research include the chemical detection module (CDM) and the enhanced chemical detection module (eCDM). The low-power corona discharge ionization and compact dual drift tube architecture of the JCAD have been the mainstay for remotely deployed miniaturized chemical vapor sensing for over a decade, going back to the DTRA Global Strike Program for Near Real-Time Battle Damage Assessment initiative. In that case, a streamlined variant of the IMS sensor was fitted to a compact module that would be projected into a region of interest to assess the potential downwind hazard from a suspected chemical weapons storage or production facility following a kinetic attack. The CDM (Figure 18) has since found application as a remote chemical vapor sensor integrated onto unmanned systems, notably under the Integrated Early Warning demonstrations sponsored by the DTRA Joint Science and Technology Office (JSTO). The CDM is a significantly modified M4 JCAD, both of which are manufactured by Smiths Detection (Edgewood, MD). The CDM is 5.72 cm (2.25 in.) in diameter and 5.72 cm (2.25 in.) high, resulting in a volume of 147 cm³. The CDM weighs 170 g (6 oz) and maintains all of the chemical detection, identification, and reporting performance capabilities of the M4 JCAD. The CDM was originally designed for a single-use application under the Global Strike initiative; therefore, it did not incorporate features to make it easy to maintain or calibrate with repeated use.

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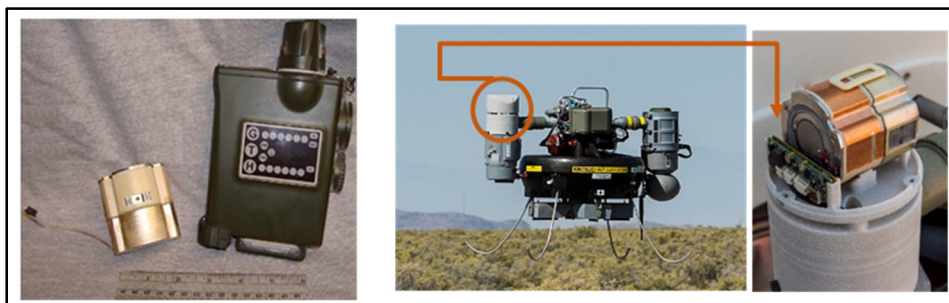


Figure 18. (U//FOUO) Left: CDM compared to the JCAD. Center and right: CDM integrated onto the T-Hawk avionics pod (Honeywell; Charlotte, NC). The CDM is functionally identical to the M4 JCAD, but is not a production component.

(U//FOUO) The C-SIRP program invests in the modernization of the CDM to improve module manufacturability and condition-based maintenance features. This updated CDM, the enhanced CDM or eCDM (Figure 19), is the most mature and established compact chemical vapor sensing option, by virtue of its atmospheric pressure ionization and ion mobility drift tube architecture. These functional components are essentially identical to the inner workings of the JCAD.

(U//FOUO)

Design changes in the eCDM relative to the legacy CDM include increased molecular sieve volume, improved data processing speed, addition of a data storage capability, increased reliability, and increased survivability. The eCDM has a projected volume of approximately 275 cm³ and a projected mass of approximately 260 g (9 oz). Delivery of 10 prototype eCDMs was scheduled for May 2020. The key disadvantage of the CDM and eCDM is their high production cost. The eCDM manufacturer, Smiths Detection, has entered into discussions with Physical Sciences, Inc. (PSI; Andover, MA) about approaches for updating the detection logic in the eCDM to improve its reliability and sensitivity while reducing false alarms. The advanced onboard detection logic for the JCAD and its COTS counterpart, the Lightweight Chemical Detector (Smiths Detection), were developed for the DARPA ChemSIGMA pilot project and SIGMA+ programs, and may be transitioned to the Compact Vapor Chemical Agent Detector program for further assessment. Details of the bona fides of the PSI detection logic are discussed in a companion technical report.³⁶

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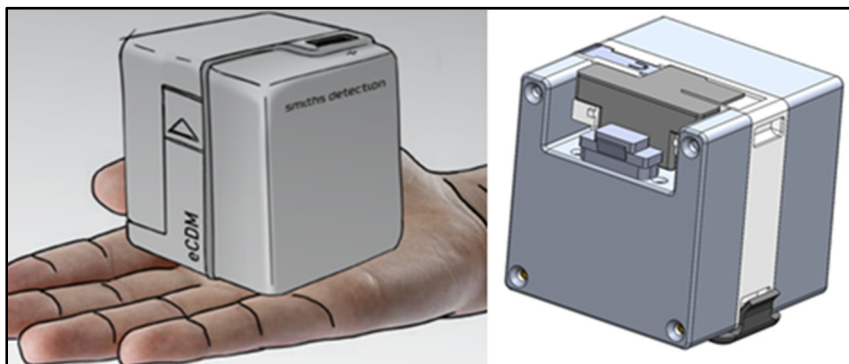


Figure 19. (U) Conceptual drawings of eCDM.

(U) Multiple firms have developed IMS sensor systems, signature libraries, and algorithms for the detection of chemical agent vapors. Most available products are as large as or larger than the JCAD, and many employ a radioisotope as an ionization source. In brief, Environics (Round Rock, TX) offers a portable/handheld system under the trade name ChemPro100i that employs an ²⁴¹Am ionization source and weighs under 2 lb. AirSense Analytics (Schwerin, Germany) offers a relatively compact IMS called the GDA-P (Gas Detector Array - Personal) that employs a ⁶³Ni ionization source and weighs about 3 lb. It is complemented by either an electrochemical cell or a photoionization detector to augment its responsiveness to certain volatile organic compounds. Chemring (Hampshire, UK) offers the Juno differential mobility spectrometer, which weighs about 2 lb.

(U) In addition to the established IMS technologies and techniques, many emerging techniques are being developed at CCDC CBC and by commercial entities. These include the incorporation of additive manufacturing (AM; also known as 3D printing). 3D printing is the process of successively and selectively adding material layer by layer, based on a computer-aided design model, to create a final object. The most common form of 3D printing is fused deposition modeling, which is an extrusion-based technique whereby molten material is

(U)

deposited from a heated extruder nozzle. The molten material is most often some formulation of plastic filament, and many filaments have specialized aesthetic or functional properties, such as wood grain appearance or electrical conductivity. Hollerbach et al. 3D printed individual conductive electrodes for use in a stacked-ring assembly IMS drift tube,^{37,38} and CCDC CBC has 3D printed sections of a unibody drift cell featuring fused layers of conductive and nonconductive material.³⁹

(U)

Current work at CCDC CBC is focused on the manufacture of an IMS-based detector with a 3D printed unibody drift cell, from ion source interface to ion collector, in a single continuous printing process that incorporates minimal assembly, size, weight, power, and cost. 3D printing offers the potential to create or replace a detector and its parts on demand from various materials at the site of need, and parts can be designed in a modular fashion to be more easily replaced on-site. Materials used in 3D printing are also inexpensive enough to warrant use in a disposable detector. A conceptual drawing of how 3D printing might be used to create critical components for a disposable IMS-based detector is shown in Figure 20.

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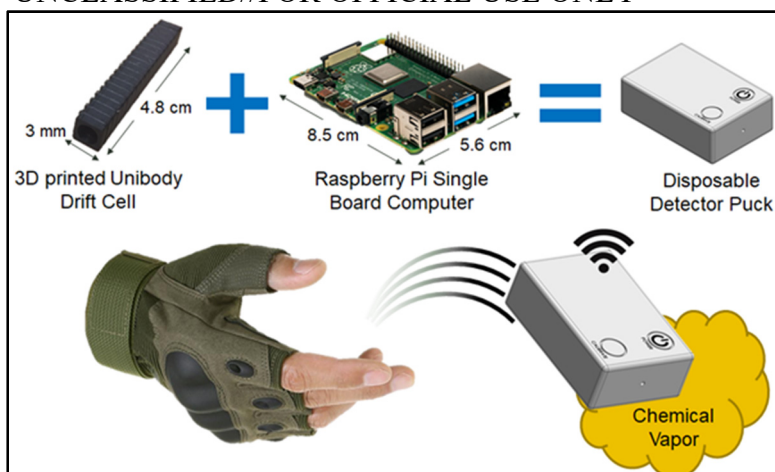


Figure 20. (U) Conceptual drawing showing the potential use of 3D-printed components for a disposable IMS-based chemical vapor detector.

4.3 (U) Colorimetric Detection

(U//FOUO) Colorimetric is a somewhat broad term that describes detection modalities where changes in visible optical properties are used to transduce a detection event into an observable signal. This can be changing color, appearance or disappearance of a color, or fluorescence changes in the sensing material. The most common colorimetric detection approach is the use of a dye that changes color upon interaction with the analyte of interest. This general approach is used in M8 paper, pH (litmus) paper, a wide variety of “dip-stick” tests for things such as chlorine or nitrate content in water, and even the M5 vesicant detector paint for Jeep hoods. In general, this sensing approach yields products with *very* low SWaP, and it is almost intuitively used by nonspecialists with minimal training requirements. Unfortunately, the downsides tend to be a lack of specificity and a lack of sensitivity. In general, colorimetric-based sensing technologies have the following advantages and disadvantages.

(U) Advantages of colorimetric-based sensing:

- (U) device simplicity can make it easily useable by generalists and specialists;
- (U) small SWaP;
- (U) inexpensive (paper reproducible on a large scale); and
- (U) response time for liquids can be rapid (<5 s).

(U) Weaknesses associated with colorimetric-based sensing:

- (U) manual reading of results can be subjective and ambiguous;
- (U) response time for vapor interaction can be long;
- (U) lack of sensitivity; and
- (U) lack of specificity.

(U//FOUO) Lack of specificity can be addressed by using multiple indicator dyes to rule out false positives. This comes at the cost of increasing complexity. If one is prepared to deal with that complexity, the “multiple indicator” approach leads to the Colorimetric Sensor Array (CSA). CSAs are small arrays printed with multiple indicator dyes, and their function is similar to the sense of smell where relatively few semi-specific receptors can discriminate between a large number of odorants by virtue of having different and characteristic patterns of activation (Figure 21).

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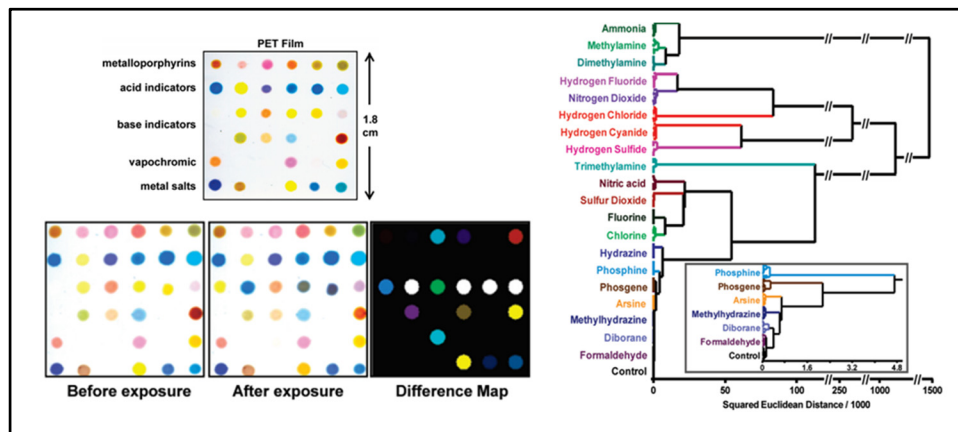


Figure 21. (U) Clockwise from top-left: a colorimetric array designed to detect TIC/TIM gases; a dendrogram showing the separation of those gases at permissible exposure limit concentration after 5 min exposures; and an example of before, after, and difference images of an exposed array. (Images from Feng et al.⁴⁰)

(U//FOUO) CSAs have been reported upon extensively⁴¹ since their invention in 2000 by Prof. Kenneth Suslick.⁴² They were studied at CDC/CBC for headspace vapor-⁴³ and liquid-phase detection of CWAs.⁴⁴ This work led to the development of a custom array type for liquid samples and a prototype optical reader (the VK3 device).

(U//FOUO) Sensitivity remains challenging, at least in the concentration regimes that are relevant for CWA vapor hazard levels. Refereed vapor testing of CSAs against isopropyl methylphosphonofluoridate (GB) vapor yielded marginal responsivity at GB vapor concentrations of 10 mg/m³. Vapor sensing was never a focus of effort for this program (the objective was to analyze a sample that was co-confined with a CSA for site exploitation and assessments), so the experiments were limited to basic checks for sensitivity. However, the results indicate a low likelihood for the CSAs to serve in detect-to-warn applications.

(U//FOUO) To provide some of the specificity offered by CSAs in a human-readable form factor, CCDC CBC is developing a “paper on demand” system for DTRA CB (project CB 10716). This system includes a library of inkjet-compatible colorimetric indicators and a response database to generate colorimetric assays against a specific target and to avoid specific interferences.

(U//FOUO) Some colorimetric products have very high sensitivity, but these are generally schemes that involve some manner of amplification, either by chain reactions where a single target molecule can cause multiple indicator molecules to change state, or by the use of enzymes that provide a similar amplifying effect by virtue of their catalytic properties. An example of a highly sensitive colorimetric indicator for a nerve agent is in the M256A2 kit, which uses purified acetylcholinesterase with a color-changing substrate to indicate the presence of a nerve agent (enzymatic detection is discussed later in this section).

(U//FOUO) Despite the challenge of achieving detect-to-warn levels of sensitivity (which pose a challenge to many detection strategies), colorimetrics can be used for vapor sensing. Colorimetric reactions can be (or can be made) irreversible or very slowly reversible. In these cases, the presence of the target over time increases the magnitude of the response. These can be used for chemical dosimetry or to detect a low concentration, provided it is present for long enough to accrue a detectable response. This property is very promising for low-cost “leave behind” or perimeter defense concepts, where an immediate response to a very low concentration may be less important than being able to cover a large area without incurring concomitantly large costs.

(U//FOUO) Clear Scientific, Inc. (Cambridge, MA) developed chemistries for agent disclosure under an Army SBIR Phase II Project, A Multicomponent Indicator Ticket for Presumptive Field Identification of Hazardous Materials. The dyes (Figure 22A) were engineered for high sensitivity to liquid-phase contaminants and response to volumes as small as 1 µL. The firm also developed a sol–gel matrix for the dyes that takes up agents on exposure to vapors and liquid or solid aerosol particles.

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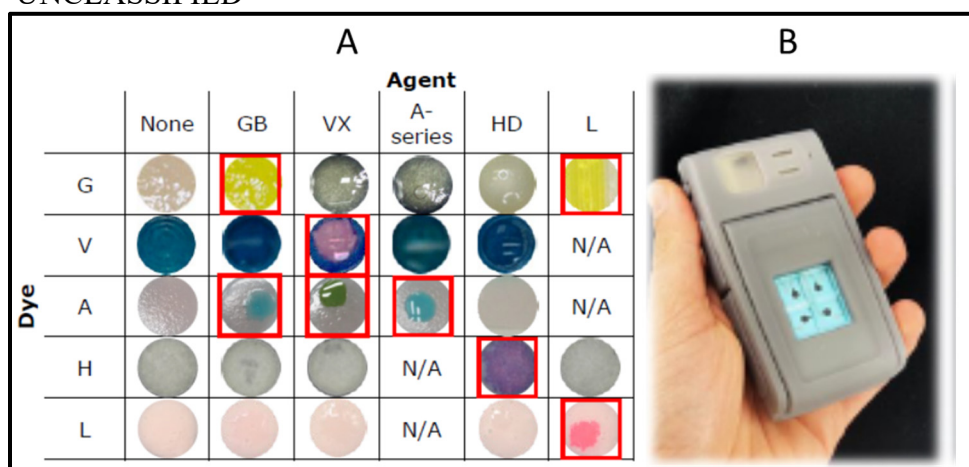


Figure 22. (U) (A) Colorimetric dyes developed by Clear Scientific, Inc. The firm embeds dyes in a sol-gel matrix that uptakes vapor, liquid, and particulate aerosols to enable air sensing applications. Note the cross-reactivity of L with the G-series dye, GB cross-reactivity, and the peculiar response of VX with the A-series dye. (B) Automated ticket reader developed by the firm. VX, *O*-ethyl-*S*-(2-diisopropylaminoethyl) methyl phosphonothiolate; HD, bis- (2-chloroethyl) sulfide; N/A, not applicable.

(U//FOUO) Clear Scientific also developed sensing technology for pharmaceutical-based agent detection under a DHS SBIR Phase II Project, Rapid Tox-Based Wearable Sensing Badge for Solid Aerosol and Contact Exposure to Fentanyl. The technology involves similar gel matrices that take in the analyte in solid form and solvate the molecules, which are then sensed via resistive monitoring of molecular-specific electrodes. The firm claims sensitivity to fentanyl and carfentanil down to 1 µg.

(U//FOUO) Intelligent Optical Systems, Inc. (Torrance, CA; <https://www.intopsys.com>) developed a multi-analyte optical sensor array (MOSA) brassboard, described as an optoelectronic chemical sensing platform for the remote detection of CWA and TIC vapors. The MOSA platform is capable of short-term, short-range broadcast of vapor detection data via WiFi that is easily interpretable with minimal end user training (Figure 23). The firm claims vapor detection performance comparable to that of a photoionization detector; however, only limited independent quantitative performance assessment data are available for the platform.

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Figure 23. (U) MOSA, developed by Intelligent Optical Systems, Inc.

(U//FOUO) An MIT-Lincoln Laboratory (LL) team is exploring the concept of a miniature fiber sensor in which a linear array of colorimetric sensing elements is set transverse to the length of an optical fiber and monitored by reflectance.⁴⁵ The low-SWaP sensors can be integrated into the fabric of a garment worn by operators or distributed as an array of unattended point sensors. Linked to a command and control system, the sensors provide early warning on exposure to a hazard agent to which the colorimetric chemistry is sensitive (Figure 24).

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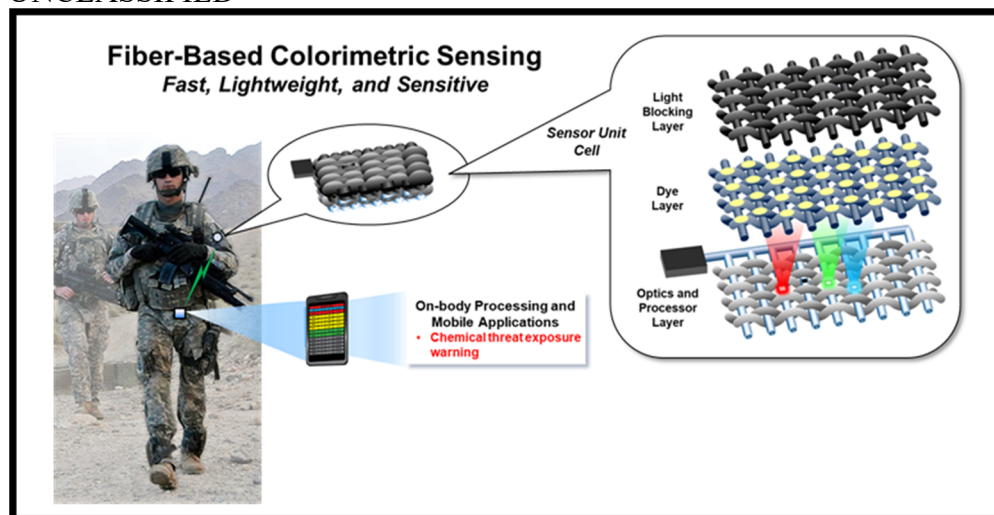


Figure 24. (U) Fiber-based sensor concept consisting of colorimetric dyes that are read by fiber arrays in an early-concept exploration project at MIT-LL.

(U//FOUO) Enzymatic reader technology is discussed here (within the colorimetric section) by virtue of its historical reduction to practice: colorimetric transduction is used in the M256 kit, which affords operationally relevant sensitivity and selectivity.

(U//FOUO) Enzymatic indication chemistries are also integrated into the Agentase Disclosure Spray technology (FLIR Systems; Wilsonville, OR). A proprietary mixture of enzymes and dyes respond to the action of agents on the enzyme by inhibiting the response, which is indicated by the dye chemistries. In the case of the HD/blister agent disclosure spray, the orange formulation changes to yellow when no HD is present, but remains orange when HD is present. In the case of the nerve agent disclosure spray, the yellow formulation remains yellow if no nerve agent is present, but turns bright red in the presence of GD or VX.⁴⁶ Little if any testing has been performed on these formulations with respect to vapor exposure; they have been evaluated principally for the intended purpose of disclosing the presence of condensed-phase contaminants on surfaces.

(U//FOUO) Although the enzymatic chemistries have largely been reduced to practice by exploiting the effect of agents on a colorimetric reaction, the use of enzymatic sensing elements in electrochemical cells is also an established technology with many applications. The readout mechanism for these electrochemical cells is readily engineered into miniature components with replaceable reagent strips.

4.4 (U) Spectroscopic Techniques (Vibrational Spectroscopy)

(U//FOUO) Optical vibrational spectroscopic techniques such as IR and Raman spectroscopy have been widely employed in various chemical biological explosive mission sets, from bulk powder or liquid detection (Raman and IR) to monitoring aerosols and vapors in the environment (IR only). Although IR aerosol and vapor systems have traditionally been large and involved logistical burdens, such as power requirements or the need for bistatic arrangements (the need for transmitter and receiver to be separately located), recent advances in both active- and passive-IR-based systems have greatly reduced these burdens. A larger discussion about IR sensors is provided in a companion technical report CCDC CBC-TR-1688.³⁶ In general, optical-based sensing technologies have the following advantages and disadvantages.

(U) Advantages of optical-based sensing technologies:

- (U) potential for small SWaP;
- (U) specificity (informational content from vibrational bands of target analyte can add greater confidence for detection); and
- (U) tunability (broadly tunable quantum cascade lasers allow for wide range of target acquisitions).

(U) Weaknesses associated with optical-based sensing technologies:

- (U) complex manufacturing/industry techniques in development;
- (U) response time for vapor interaction can be long (sorbent technologies being developed to decrease response time);
- (U) low sensitivity; and
- (U) lower maturity.

(U//FOUO) Raman-based systems have traditionally had minimal to no capability in vapor- and gas-phase sensing due to the combination of weak Raman scattering cross sections and dilute molecular concentrations. One previous example was the Chemical Agent Raman Detection (CARD) research program funded by the DTRA JSTO for Chemical and Biological Defense (Figure 25). Developed between 2013 and 2016, the CARD system successfully detected various aerosolized CWAs for real-time monitoring.⁴⁷ Other large benchtop systems that utilize a long optical path length⁴⁸ or hollow-core optical fibers⁴⁹ have been demonstrated, but the development of a sensitive, handheld Raman system for trace vapors has proven challenging.

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Figure 25. (U) Prototype CARD system developed for Raman-based vapor detection.

(U//FOUO) Recent advancements in silicon photonics capitalize on dramatic miniaturization, mimicking those that electronic ICs experienced decades ago as it transitioned from assembled discrete components to modern chip ICs. This newly established capability combines modern silicon processing and breakthrough optical component designs to make complex photonic integrated circuits (PICs) on a single substrate. The telecommunications industry and advanced computing drive much of the PIC innovation; however, this technology has great potential to advance DoD applications. Placing or fabricating prealigned functional optical components (i.e., lasers, detectors, spectrometers, etc.) onto a single chip could enable cheap, low-SWaP multiplexed sensor arrays and spectrometers on a chip and revolutionize CB threat detection.

(U//FOUO) Waveguide-enhanced Raman spectroscopy (WERS) is a promising technique for significantly increasing the signal that can be obtained from typically weak Raman scattering processes. In WERS, the laser pump light is confined in a single-mode waveguide over a much longer interaction length than is typically encountered in conventional Raman spectroscopy.^{50–52} Molecules in the evanescent field of the waveguide generate Raman scattered light that can be efficiently coupled back into the waveguide. The long interaction path length sharply increases the signal levels as compared to conventional Raman microscopy. If a sorbent coating is deposited on the waveguide surface, then vapor-phase chemicals concentrate in the waveguide's evanescent field, further enhancing trace detection.

(U//FOUO) Typically, WERS measurements are performed using benchtop, research-grade laboratory Raman spectrometers with deeply cooled detectors to increase sensitivity and decrease noise. However, manufacturers have recently introduced compact, even handheld, Raman spectrometers that have continuously improving sensitivity.⁵³ By integrating a sorbent-coated optical waveguide produced on a PIC chip with a compact Raman spectrometer, development of a handheld Raman sensor for vapor detection may be possible. This would be useful for a variety of applications, including detection of hazardous vapors such as CWAs and volatile components of explosives and narcotics. For example, compact 785 and 1064 nm Raman spectrometers from Wasatch Photonics (Logan, UT) are being used to perform WERS measurements. In work supported by DTRA and the USA, the WERS technique is being developed to increase its sensitivity and move toward a handheld WERS-based Raman sensor (Figure 26). Challenges include reducing waveguide fluorescence, developing novel sorbent polymers with fast absorption kinetics that can be uniformly coated on PIC chips, reducing waveguide losses, and incorporating additional components such as spectral filters directly on a chip.⁵⁴

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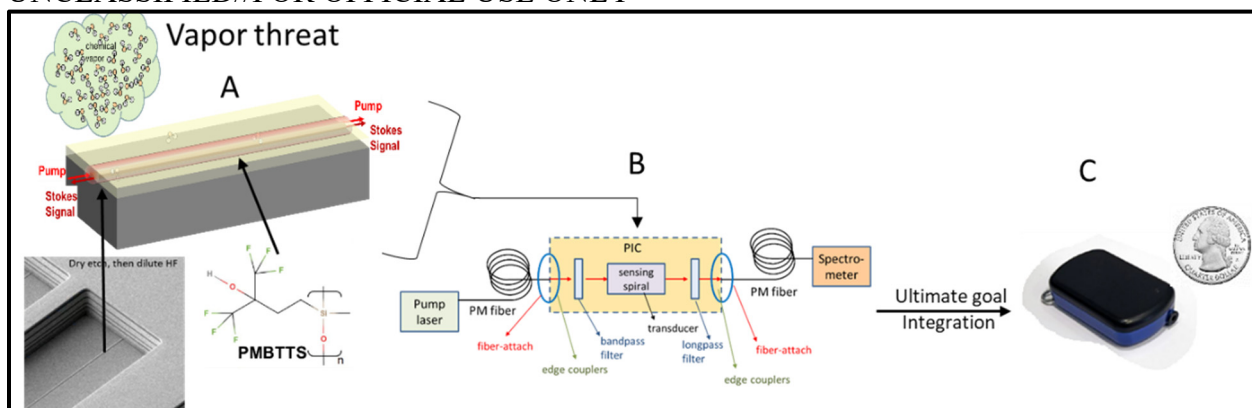


Figure 26. (U) (A) Waveguide schematic shows sorbent material for capturing target vapor threat. (B) Schematic diagram of setup used to perform WERS measurements with compact Raman spectrometers. (C) Technology integration goal for optical-based vapor sensing into compact Raman spectrometer (BaySpec [San Jose, CA] mini Raman spectrometer, shown as an example of a small compact Raman device and not necessarily the target final form factor).

(U//FOUO) Pendar Technologies (Cambridge, MA; <https://www.pendartechonologies.com>), in partnership with Mulberry Sensors (Cambridge, MA; <https://mulberrysensors.com>), developed low-cost miniature IR sensors that exploit the narrow emission linewidth of single-mode, distributed-feedback quantum cascade lasers and silicon optical-fiber waveguides fabricated in a compact and relatively inexpensive form factor. Prototype chip-scale sensors (Figure 27) have been demonstrated for parts-per-million-level detection of ethylene, methane, carbon dioxide, carbon monoxide, acetylene, and nitrous oxide. Each sensor (estimated to cost ~\$250 in small-scale production) is dedicated to the detection of a single analyte.

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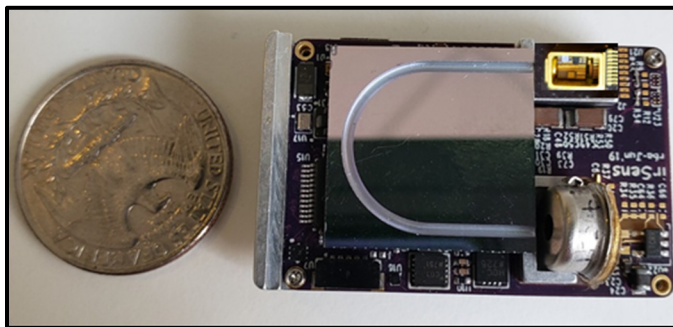


Figure 27. (U) Chip-scale distributed-feedback quantum cascade laser sensor developed by Pendar Technologies and integrated by Mulberry Sensors.

4.5 (U) Additional Miniature Sensing Concepts

(U//FOUO) A team at MIT-LL explored an ultra-low SWaP chemical sensor that embeds a colony of bioluminescent bacteria onto the surface of a microchip assembly fitted with an avalanche photodiode. The selected bacteria luminesce when exposed to a toxic environmental stressor. The team experimented with microbial species capable of surviving in a lyophilized state and exhibiting sensitivity to threat vapors once “activated”. Their prototype assembly demonstrates the concept (Figure 28).

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Figure 28. (U) Microsensor prototype consisting of bioluminescent bacteria embedded in a microchip construct (left) and housed in a miniature sensor sphere (right) in an early concept exploration project at MIT-LL.

5. (U) SUMMARY

(U) This report summarizes a series of LIPCS concept of employment discussions with operational end users for each of the Services as well as USSOCOM. Candidate technologies at various maturity levels were considered for the LIPCS application (Figure 29). Upon surveying these technologies, we conclude that at present, the availability of GOTS and COTS solutions that fit into the LIPCS desired format and functionality is low. All technologies will require additional maturity; however, impedance-based sensing techniques (including graphene sensing, MOFs, MIPs, functionalized carbon nanotubes, etc.) exhibit the most probable near-term S&T solutions for vapor sensing performance and reliability within the SWaP form factor desired for the LIPCS concept. Colorimetric chemistries may offer a near-term solution for persistent surface contamination monitoring applications.

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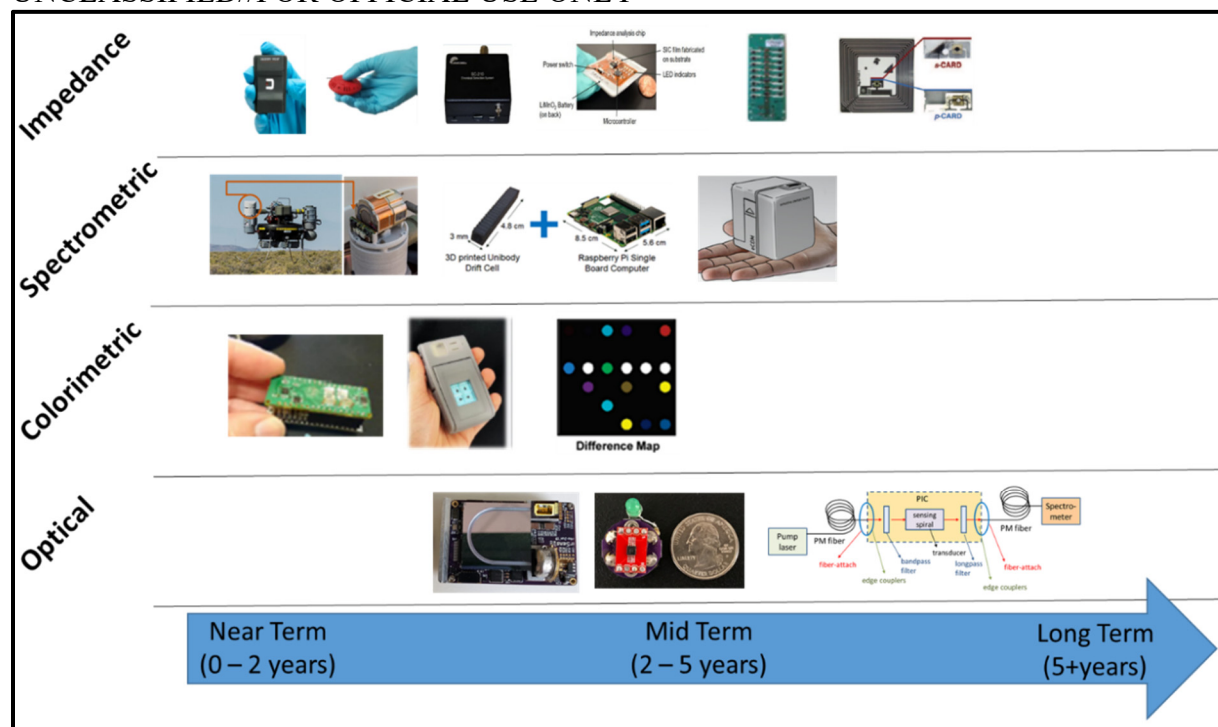


Figure 29. (U) Surveyed technologies for LIPCS solutions. Probable timelines are shown for S&T development and prototype availability and assessment.

(U//FOUO) Furthermore, these technologies are generally point sensors that sample the environment to generate a response. Some LIPCS use cases involve detection of low-volatility and/or persistent chemical agent on a surface; therefore, the sensor must interact with surfaces by direct contact. Alternatively, the LIPCSs may be useful when pre-positioned by R&S teams prior to an adversary's use of a persistent threat agent to contaminate an area and thereby deny access or cause maneuver forces to encounter the threat agent and become contaminated. As technology development continues toward the leave-in-place concept, the need for proper sampling must fully inform sensor design.

(U//FOUO) *Impedance Technologies.* The reviewed impedance-based chemical sensors provide near-term (0–2 years) opportunities to fulfill the LIPCS role. These devices can be small and inexpensive, and they have potential to provide anomalous threat detection of chemical hazards. They are in various states of internal and commercial investigation. The main questions moving forward with impedance-based sensors are related to sensitivity and specificity, to ensure that alarm levels will not be hindered by background interference and environmental effects. Chemicapacitor technologies consisting of functionalized sensing elements with different chemical affinities performed impressively as reported in recent literature.

(U//FOUO) *Spectrometric Technologies.* IMS remains the most mature technology for point chemical sensing. However, developments in miniaturization must still address the need to reduce false-alarm rates that are typical with fielded IMS devices. In addition, IMS systems carry higher SWaP and cost than are desirable for the LIPCS concept.

(U//FOUO) *Colorimetric Technologies.* Colorimetric technology is a mainstay for reliable, inexpensive sensing of liquid-phase contamination, but it requires direct contact between the indicator dyes and the target analyte. Colorimetric detection of vapor-phase threat agents is possible and can be exquisitely sensitive when enzymatic chemistries are incorporated into the sensing element; however, many of these chemical reactions are slow, leading to longer-than-desirable response times. Digital transduction mechanisms involving diode illumination and detection or fiber optics provide an array of options for the automated reporting function. The best use case for colorimetric technologies, including self-indicating colorimetric transduction mechanisms and fiber-based systems, would be in situations where the sensors were deployed in advance of an adversary's use of a persistent surface-contaminating threat agent.

(U//FOUO) *Optical Technologies.* Optical technologies may provide greater specificity and sensitivity than the other technological classes, but the examples discussed here require far more development. In the case of PICs and WERS, the use of hypersorbent polymers to concentrate and capture analytes could provide highly specific chemical sensors to vapor threats in the mid to long term.

6. (U) CONCLUSION AND PATH FORWARD

(U//FOUO) This report captures the military operational concepts for LIPCS and the most viable sensor development candidates for LIPCS applications in the near term and beyond. From discussions with representatives from the main Service branches, we conclude that although employment concepts still need to be fleshed out, LIPCS is a desirable technology concept that does not currently exist. Information gathered from the surveyed technologies appears to demonstrate that viable solutions are in various states of maturity. Technologies either need to be assessed to establish a baseline performance against current targeted threats, which will determine whether further development is warranted, or they are already being invested in and require additional time and/or investment to mature. We have also determined that two variants of the LIPCS would be needed to address two distinct chemical threat circumstances of vapor and condensed chemical threats.

(U//FOUO) We conclude that impedance-based sensors (including chemiresistors based on functionalized graphene or other 2D materials) and enzyme-catalyzed colorimetric chemistries are the most capable near-term technologies for meeting the LIPCS concept while remaining within the SWaP and unit-cost constraints. However, the technology currently on the market has low maturity with respect to CWA detection. The microcapacitance sensor technology affords the greatest potential for achieving relevant sensitivity and specificity while accommodating background clutter. The colorimetric and colorimetric impedance-based technologies (such as the Intelligent Optical Systems MOSA device and the self-indicating colorimetric systems) are best suited to detection of surface-contaminating condensed-phase threats and can be assessed for suitability in this application in the near term. Table 3 outlines the two groups of technologies for investigation and maturation.

Table 3. (U) LIPCS Recommendation for Assessment
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Priority	Sensor Technology	Path Forward
LIPCS Vapor Sensor	Carbon nanotube technologies	Inquire into the current availability of developmental prototypes from industry; obtain and assess against select chemical vapor threats for baseline performance
	Microchemicapacitance technologies	Longer-term R&D projects to develop highly sensitive vapor sensing arrays
LIPCS Surface Contamination Sensor	Self-indicating colorimetric/capacitance (CCDC CBC)	Assess material chemistries from multiple sources for side-by-side performance comparison against persistent chemical threats (condensed phase)

(U//FOUO) Colorimetric chemistries are more mature than impedance-based approaches; however, they are most effective in direct sensing of condensed-phase (liquid droplet) threats, making them more suitable for the detection of an adversary's use of a persistent chemical hazard to contaminate surfaces. We note that the technology should be investigated in terms of aerial contamination sensitivity and response time and for detection of persistent or low-volatility agents on surfaces.

(U//FOUO) The LIPCS concept would be best implemented as a complementary suite of inexpensive, small, lightweight sensors, with some engineered for optimal performance as vapor sensors and others for detection of surface contamination. Longer-term S&T development includes the further exploration of miniature ion mobility devices and optical technologies such as integrated photonic applications including WERS and spectrometer-on-a-chip.

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(U) ACRONYMS AND ABBREVIATIONS

(U)	2D	two dimensional
(U)	3D	three dimensional
(U)	AA	avenue of approach
(U)	AVCAD	Aerosol and Vapor Chemical Agent Detector
(U)	CARD	Chemical Agent Raman Detection
(U)	CB	chemical and biological
(U)	CBOA-2019	2019 Chemical Biological Operational Analysis event
(U)	CBRN	chemical, biological, radiological, and nuclear
(U)	CCDC CBC	U.S. Army Combat Capabilities Development Command Chemical Biological Center
(U)	CCDC Soldier Center	U.S. Army Combat Capabilities Development Command Soldier Center
(U)	CDM	chemical detection module
(U)	CONEMPS	concepts of employment
(U)	COTS	commercial off the shelf
(U)	C-SIRP	CBRN Sensors Integrated on Robotic Platforms
(U)	CNT	carbon nanotubes
(U)	CWA	chemical warfare agent
(U)	DARPA	Defense Advanced Research Projects Agency
(U)	DHS	Department of Homeland Security
(U)	DTRA	Defense Threat Reduction Agency
(U)	eCDM	enhanced chemical detection module
(U)	EOD	explosive ordinance disposal
(U)	EUD	end user device
(U)	GB	isopropyl methylphosphonofluoridate
(U)	GOTS	government off the shelf
(U)	HFIP	hexafluoroisopropanol
(U)	IC	integrated circuit
(U)	IDC	interdigitated capacitors
(U)	IDE	interdigitated electrode
(U)	IDLH	immediately dangerous to life or health
(U)	IMS	ion mobility spectrometry
(U)	IR	infrared
(U)	ISR	intelligence, surveillance, and reconnaissance
(U)	JCAD	Joint Chemical Agent Detector
(U)	JSTO	Joint Science and Technology Office
(U)	LIPCS	leave-in-place chemical sensor
(U)	LL	Lincoln Laboratory
(U)	LP/OP	listening or observational post
(U)	MIP	molecular imprinted polymer
(U)	MIT	Massachusetts Institute of Technology
(U)	MOF	metal-organic framework
(U)	MOSA	multi-analyte optical sensor array

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(U)	MSR	main supply route
(U)	NAI	named area of interest
(U)	NBCRV SSU	Nuclear Biological Chemical Reconnaissance Vehicle Sensor Suite Upgrade
(U)	PIC	photonic integrated circuit
(U)	PSI	Physical Sciences, Inc.
(U)	R&S	reconnaissance and surveillance
(U)	RADAS	Rapid Airfield Damage Assessment System
(U)	RF	radiofrequency
(U)	RFID	radiofrequency identification
(U)	S&T	science and technology
(U)	SBIR	Small Business Innovation Research
(U)	SEAL	sea, air, and land
(U)	SIC	self-indicating colorimetrics
(U)	SWaP	size, weight, and power
(U)	TAI	targeted area of interest
(U)	TIC	toxic industrial chemical
(U)	TIM	toxic industrial material
(U)	TRSS	Tactical Remote Sensor System
(U)	UAS	unmanned aircraft system
(U)	UGS	unattended ground sensor
(U)	UGV	unmanned ground vehicle
(U)	USA	U.S. Army
(U)	USAF	U.S. Air Force
(U)	USMC	U.S. Marine Corps
(U)	USN	U.S. Navy
(U)	USSOCOM	U.S. Special Operations Command
(U)	VBSS	visit, board, search, and seizure
(U)	WERS	waveguide-enhanced Raman spectroscopy
(U)	WMD	weapons of mass destruction

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